BHARAT101_2G_USER_MANUAL_V2.8

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Contents

1 Introduction 9
1.1 Safety instruction 9
1.2 Copyright Notice 9
2 About the Document 9
3 Acronyms & Definitions 9
4 Basic Description 10
5 Inside the Box 10
6 Basic Characteristics 10
6.1 GSM/GPRS 10
6.2 GNSS Module 10
6.3 Hardware features 11
6.4 Interface features 11
6.5 Firmware Features 11
6.6 Mechanical and Environmental specification. 11
6.7 Operating Condition 11
7 Internal Battery 12
7.1 Battery Disposal Instruction: 12
8 Certification 12
9 Mechanical Overlay 12
10 Connector Pinout from Device End 13
11 LED Indications 13
11.1 Power LED (Red) 13
11.2 GNSS LED (Blue) 13
11.3 Process LED (Red) 13
11.4 GSM LED (Green) 14
12 Configuration 14
12.1 System requirements 14
12.2 Installation of Driver 14
12.3 Firmware Update 15
12.3.1 Using Flash Tool 15
12.3.2 Upgrade Procedure 15
13 Settings and Configuration 20
13.1 Command Format 20

13.1.1 set 20
13.1.2 get 23
13.1.3 clr 25
13.1.4 debug 27
13.1.5 diag 28
13.1.6 Header Errors 29
13.1.7 Source List 29
14 Config 360 30
14.1 System 32
14.1.1 GNSS 32
14.1.2 Time Offset 35
14.1.3 Battery Charging mode 35
14.1.4 Covert mode 36
14.1.5 Movement Detection 36
14.1.6 Ignition Detection 37
14.1.7 Vehicle Number 39
14.2 Network 39
14.2.1 APN 40
14.2.2 Primary Profile Name 40
14.2.3 Primary profile APN 40
14.2.4 Secondary profile name 41
14.2.5 Secondary profile APN 41
14.3 Communication 41
14.3.1 Primary server IP/Domain 42
14.3.2 Primary server Port 42
14.3.3 Secondary server IP/Domain 42
14.3.4 Secondary server Port 42
14.3.5 Packet Acknowledgement Timeout 43
14.3.6 Common Layer 43
14.4 Emergency 43
14.4.1 Emergency server IP/Domain 44
14.4.2 Emergency server port 44
14.4.3 Emergency Recording Interval 45
14.4.4 Emergency Auto Clear Time 45
14.4.5 SMS Fallback Interval. 45
14.4.6 Control Centre Number 45

14.4.7 Emergency Clear 46
14.5 Data Recording 46
14.5.1 Time-based recording 46
14.5.2 Ignition Off Interval for Primary 46
14.5.3 Ignition On Interval for Secondary 47
14.5.4 Ignition Off Interval for Secondary 47
14.5.5 Heading Based Recording 47
14.5.6 Distance-Based Recording 48
14.5.7 Login packet Header 48
14.5.8 Tracking Packet Header 49
14.5.9 Health Packet 49
14.5.10 Command Response Header 49
14.5.11 Datatype Change (Heading and Odometer) 50
14.6 Eco Driving 50
14.6.1 Eco Driving Source 51
14.6.2 Eco Driving Axis 51
14.6.3 Harsh Acceleration 51
14.6.4 Harsh Braking 53
14.6.5 Harsh Cornering 55
14.6.6 Over speed 56
14.6.7 Accident Detection 58
14.7 Sleep Mode 59
14.7.1 User Commands 59
14.8 IO 60
14.8.1 Digital Inputs 60
14.8.2 Analog Inputs 62
14.8.3 Digital Output 66
14.9 Security 67
14.9.1 Change Password 68
14.9.2 Mobile Number whitelisting 68
14.10 Maintenance 70
14.10.1 Maintenance Service 70
14.10.2 FOTA 72
14.11 Miscellaneous 75
14.11.1 Tilt/Fall 75
14.12 UART/RS232 77

14.12.1 User Command 77
14.12.2 UART 77
14.12.3 ECU 91
14.13 Restricted Vehicle Access 95
14.13.1 Identification 95
14.13.2 Authorisation 95
14.13.3 User Commands 95
15 Debug Commands 96
15.1 Switch Network 96
15.1.1 User command 96
15.2 Forced Restart 96
15.2.1 User Command 96
15.3 Activation Command 97
15.4 HCHK Command 97
15.5 Get All 98
16 Diagnostic commands: 102
16.1 General Device Information 103
16.2 Get Location link 103
16.3 IO Status 104
16.4 Get all firmware versions 104
16.5 Get GNSS Information 105
16.6 Get GSM Information 105
16.7 deviceinfo 105
17 Bluetooth Application 106
17.1 Mobile Requirements 106
17.2 Log in to the Mobile App 106
17.3 Scan And Connect 107
17.4 Password Verification 108
17.5 Application Features 109
17.6 Device Info 110
17.7 GNSS Info 110
17.8 GSM Info 111
17.9 Input/Output 112
17.10 Configuration 112
17.11 Debug Mode 113
17.12 Quick Mode 114

17.13 Manual Mode 114
17.14 Logging Mode 115
17.14.1 Logout/Disconnect 115
18 Device installation 116
18.1 Preparation for the Installation 116
18.1.1 Pre Installation notes 116
18.1.2 Installation Safety 116
18.1.3 Tools Required 117
18.2 Installation Instructions 117
18.2.1 Identification of Installation location. 117
18.2.2 The orientation of the Device 118
18.2.3 Identify the Wires 119
18.3 Best Installation Practices 119

1 Introduction

1.1 Safety instruction

This chapter contains information on how to operate BHARAT 101 safely. By following these instructions and recommendations, risky situations can be avoided. Read the instructions carefully and follow them strictly while operating the device.

- BHARAT 101 device is designed to work in an automotive environment with a supply range of +9V to +32V DC power supply.
- To avoid mechanical damage, it is advised to transport the BHARAT 101 device in an impact-proof package.
- Connect the Device to the wiring harness only after the connection with the vehicle is complete
- Please terminate any open-ended/Skinned wires in the wiring harness with proper insulation to avoid short circuit and disturbance to the vehicle wiring harness. The manufacturer is not responsible for any damage to the Device or vehicle due to faulty wiring or installation.

1.2 Copyright Notice

The information disclosed here is the exclusive property of iTriangle Infotech Pvt ltd and no part of this document can be reproduced or transmitted in any form or by means including electronic storage, reproduction or execution without the prior written permission from iTriangle Infotech Pvt Ltd. The information contained in this document is subjected to change without notice and should not be construed as a commitment by iTriangle Infotech Pvt Ltd unless such commitment is expressly given in a separate document.

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2 About the Document

This manual covers the various features of BHARAT 101 and information about the terminologies, possibilities, electrical, mechanical characteristics & configuration of the BHARAT 101 device. The device hereafter refers to aQuila Bharat101 device.

3 Acronyms & Definitions

PC	Personal Computer
GPRS	General Packet Radio Service.
GSM	Global System for Mobile Communication.
SMS	Short Message Service
BT	Bluetooth
AC	Alternating Current
I/O	Input /Output
GNSS	Global Navigation Satellite System

DI	Digital Input
DO	Digital Output
DC	Direct Current

TABLE 1: ACRONYMS

4 Basic Description

BHARAT 101 is an electronic device that locates its position data, records its interface information and sends it by using a GPRS network. This device is suitable for applications, where it needs the location, status of the remote asset along with control of its peripherals. It is important to mention that BHARAT 101 has additional inputs, outputs, and a serial communication feature that will control and monitor other devices/status on a remote asset. It has a USB, GPRS, and SMS channel for configuration of settings and to get the status of the device. The BHARAT 101 device complies with AIS140 standards. The device also supports multiple Panic buttons which are used to generate an emergency alert in case of emergency conditions. The device comes with built-in status indicators, which show statuses like GSM modes, GNSS fix, Processing status, and power status that can be monitored easily by looking into the device. Most importantly, **BHARAT 101 is the first device in India that is certified under AIS140 compliance.**

5 Inside the Box

- BHARAT 101 device.
- Vehicle Harness Cable with 12 pin connector.
- Side Panel.
- Screws for side panel- 2Nos.
- Mounting screws-4Nos.
- Cable ties for installation-2Nos.
- Warranty card.

6 Basic Characteristics

6.1 GSM/GPRS

- Quad-Band GSM 850/900/1800/1900 MHz
- GPRS: Class 10 Coding Scheme CS1 (1W @ 1800/1900MHz) to CS4 (2W @ 850/900MHz).
- SMS (text).
- GPRS multi-slot class 12/10.

6.2 GNSS Module

- Multi GNSS constellation support with GPS and IRNSS.
- Tracking Sensitivity of -160dBm.
- Location Accuracy of 2.5 m CEP @ -130 dBm.
- Inbuilt embedded antenna.

6.3 Hardware features

- MT2503A Chipset.
- Solid State Flash, 100K Erase and Program Cycle, 10 Year data retention.
- Internal backup battery.
- UICC with the dual profile.
- GNSS back up power for a quicker fix.

6.4 Interface features

- Wide DC Input Voltage Range [9V – 32V].
- USB Port device type-B.
- 4 Digital Inputs.
- 2 Digital Outputs [Open drain].
- 2 Analog Inputs.
- 1 Serial Port [RS232].
- LEDs to indicate device status.

6.5 Firmware Features

- Detect Harsh Breaking, Harsh Acceleration, and Rash Turning/Harsh Turning.
- Firmware upgrade over the air.
- Configuration over the air (SMS, GPRS).
- OTA response alert for Parameter change.
- Frame Number for each recorded packet.
- Data acquisition – Heading, time, and distance-based.
- Safe immobilization.
- Immobilizer violation alert.
- Plug & play UART interface.
- Maintenance service.
- Tilt/Fall with Dynamic orientation.
- Multi sleep modes.
- Message-ID (Event ID) based detection mechanism for faster detection of alerts.
- Packet Type-based detection mechanism for faster detection of alerts.
- Up to 40000 tracking record storage.

6.6 Mechanical and Environmental specification.

- Dimensions (mm X mm X mm): 119.32 X 67.6 X 34.
- Weight: 209 Gram.
- Enclosure: ABS plastic casing with IP 67 rating.

6.7 Operating Condition

- Operating temperature: -25 °C to +80°C
- Storage temperature: -25 °C to +80°C
- Humidity: 95% Non-condensing

7 Internal Battery

- Li-Ion Rechargeable Battery, 3.7V, 750mAh
- Operating time with internal backup battery depends on temperature, data sending frequency, GPRS, accumulator age, and the number of charge/discharge cycles.

CAUTION: Risk of exploding if the battery is induced to mechanical damage.

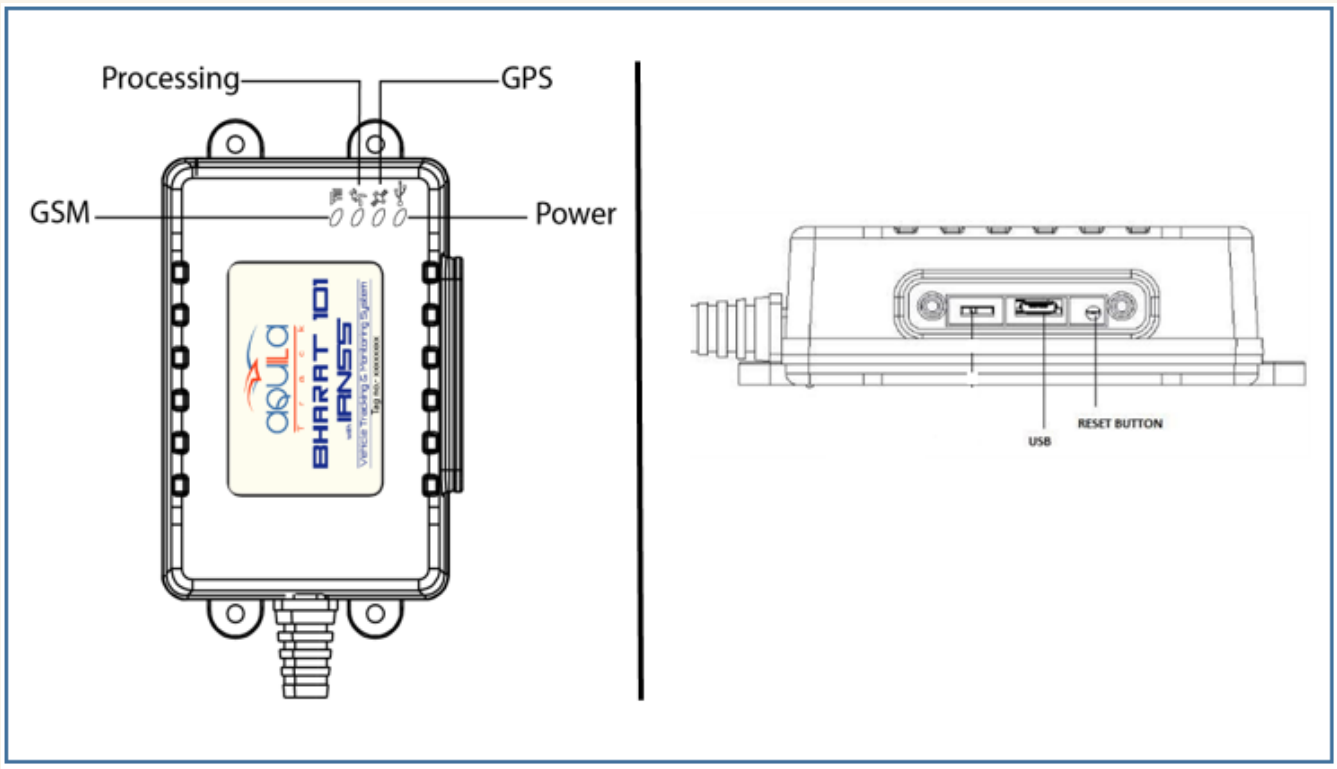
7.1 Battery Disposal Instruction:

Battery should not be disposed of with household waste. Damaged batteries are Disposed to a battery recycle bin found in stores.

8 Certification

- AIS140

9 Mechanical Overlay



10 Connector Pinout from Device End

Pin No	Wire Color	Pin Name	Description
1	Red	(+) VCC	Power Supply Pin(9 – 32) V DC
2	Black	(-) GND	(-) Ground pin
3	Light Grey	DOUT-1	Digital Output 1
4	Dark Grey	DOUT-2	Digital Output 2, [Reserved for Panic]
5	Yellow	Analog Input-1	Analog Input 1
6	Orange	RS232-RX	Serial Data Interface RX*
7	White	RS232-TX	Serial Data Interface TX*
8	Blue	DIN-2	Digital Input 2
9	Purple	DIN-1	Digital Input 1
10	Light Brown	DIN-3	Digital Input 3
11	Dark Brown	Analog Input-2	Analog Input 2
12	Green	DIN-4 (IGN)	Digital Input, (Ignition)

*The indication is with respect to the Bharat101 C device.

11 LED Indications

11.1 Power LED (Red)

Behaviour	Significance
ON	Main power is connected
OFF	Main power is not connected/ Covert Mode/Sleep Mode

11.2 GNSS LED (Blue)

Behaviour	Significance
Switched OFF	No GNSS fix/Sleep Mode/ Covert Mode
500ms on/1sec off	GNSS fix available

11.3 Process LED (Red)

Behaviour	Significance
Continuous On	Till initialization
500ms On /2sec Off	After initialization
Switch OFF	Sleep mode/ Covert Mode

11.4 GSM LED (Green)

Behaviour	Significance
Off	Not Registered/Covert Mode/ Sleep Mode
500ms On/1sec Off	Registered to the network
500ms On/2sec Off	Bearer is activated
500ms On/4sec Off	A socket connection is successful
100ms On/50ms Off	During data transmission

12 Configuration

12.1 System requirements

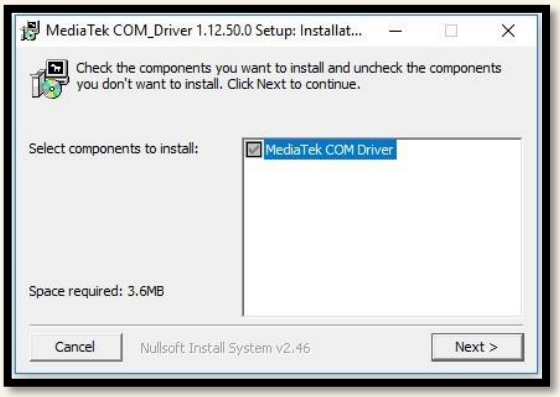
- Operating system version: 32-bit and 64-bit.
- Operating system: Windows 7 and later.
- Any serial communication tool and Micro USB type-B cable.
- A USB to Serial Cable
- A DB9 to Open-Ended wire Connector

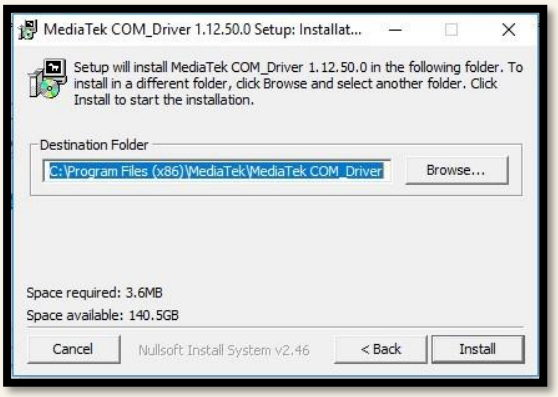
12.2 Installation of Driver

Steps to be followed to install the drivers

- Get the driver from FAE/ iTriangle support team.
- Extract the driver and install it by opening “InstallDriver.exe”

- Follow the guidelines as per the installer.





- Connect the device to the PC using a USB cable.
- Cross verify the port enumeration in Device manager→ Ports. Total 2 port should be enumerated by either of the below names.
    - USB Debug Port
    - USB Modem Port
    - USB Serial Port

12.3 Firmware Update

12.3.1 Using Flash Tool

Flashing of BHARAT 101 firmware to Device can be done by using flash Tool. iTriangle FAEs will have the flash tool. Get the flash tool and the latest firmware from FAEs. Below are the steps to update the firmware of the device. Below is the link to download the flash tool.

Click here to download the flash tool

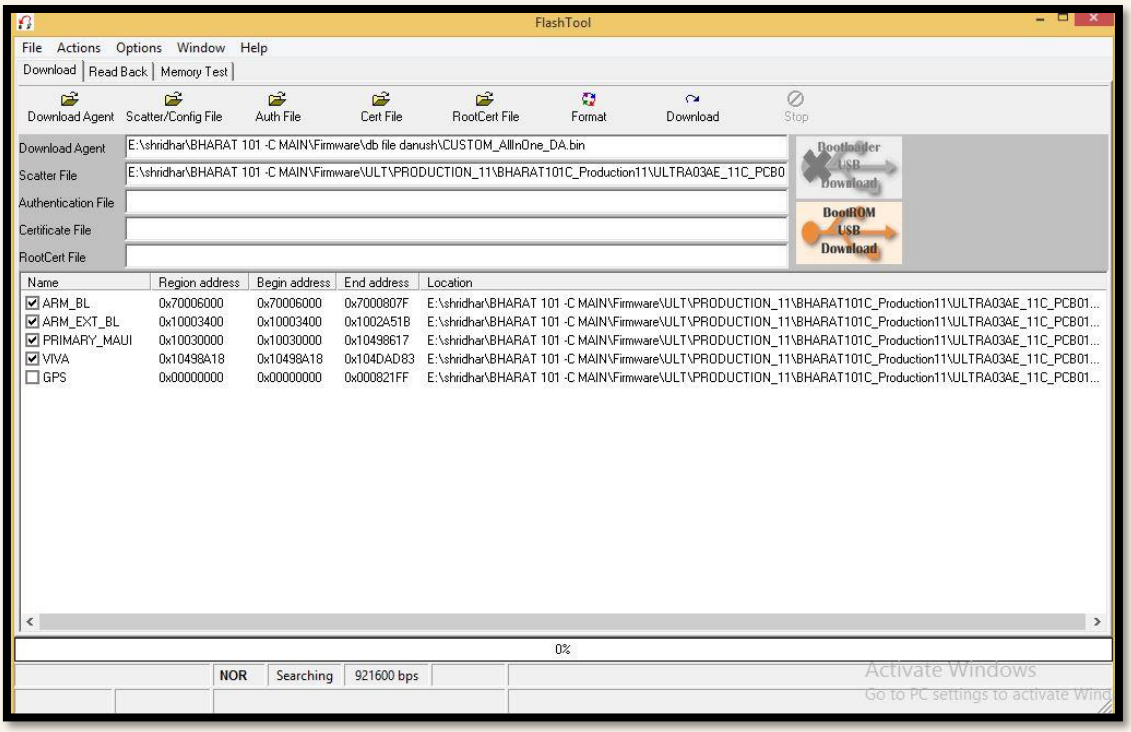
12.3.2 Upgrade Procedure

The tool is used to upgrade the firmware. It works according to the following steps.

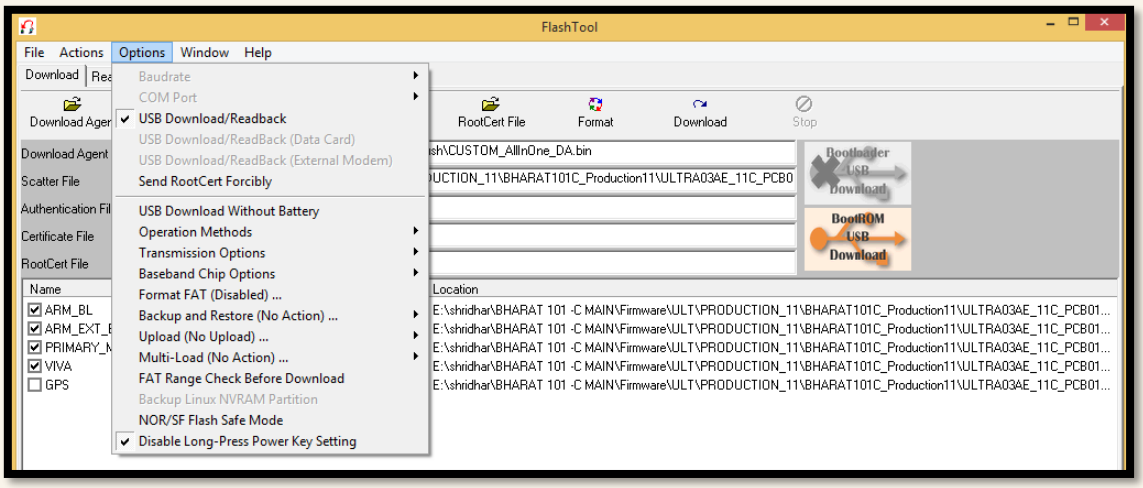
- Configuration of the tool.
- Loading Scatter file.
- Download the firmware.

12.3.2.1 Configuration of the upgrade tool

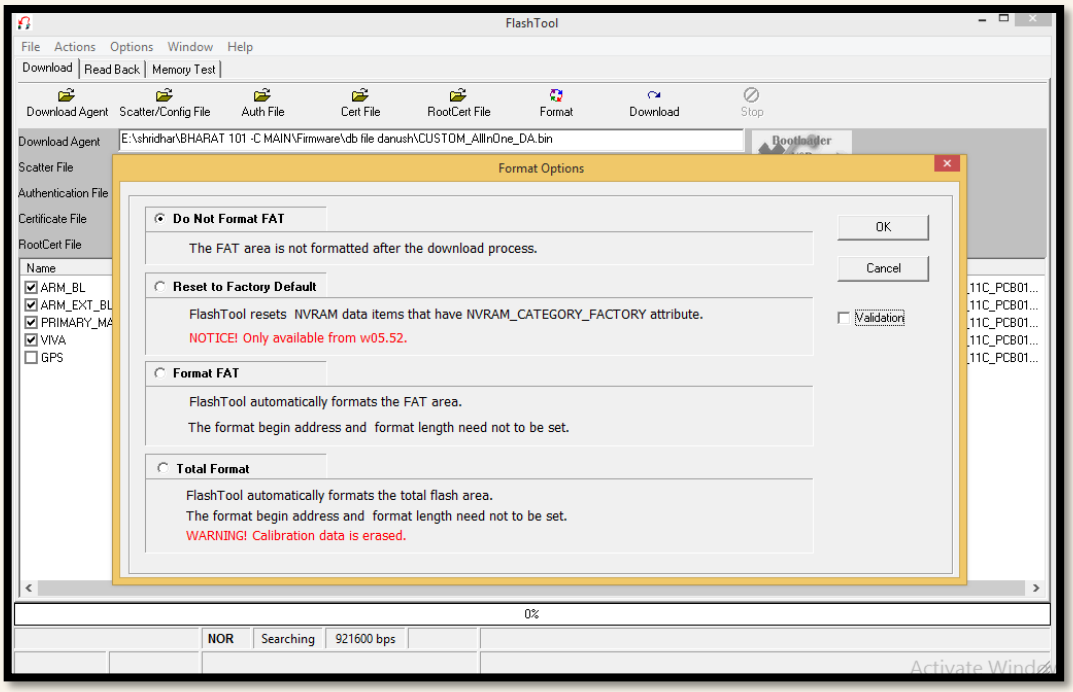
- Open the flash tool by double-clicking on the “Flash_tool” application from the directory structure “Flash_Tool_V1.0→FlashTool_v5.1640.00”. As soon as the flash tool is opened the main interface is shown below.



- Before proceeding with the flash process, go to Options and follow the procedure as follows.
  - Disable Long –Press power key setting option should be enabled.
  - USB download/Readback should be enabled.
  - Format FAT (Disabled)



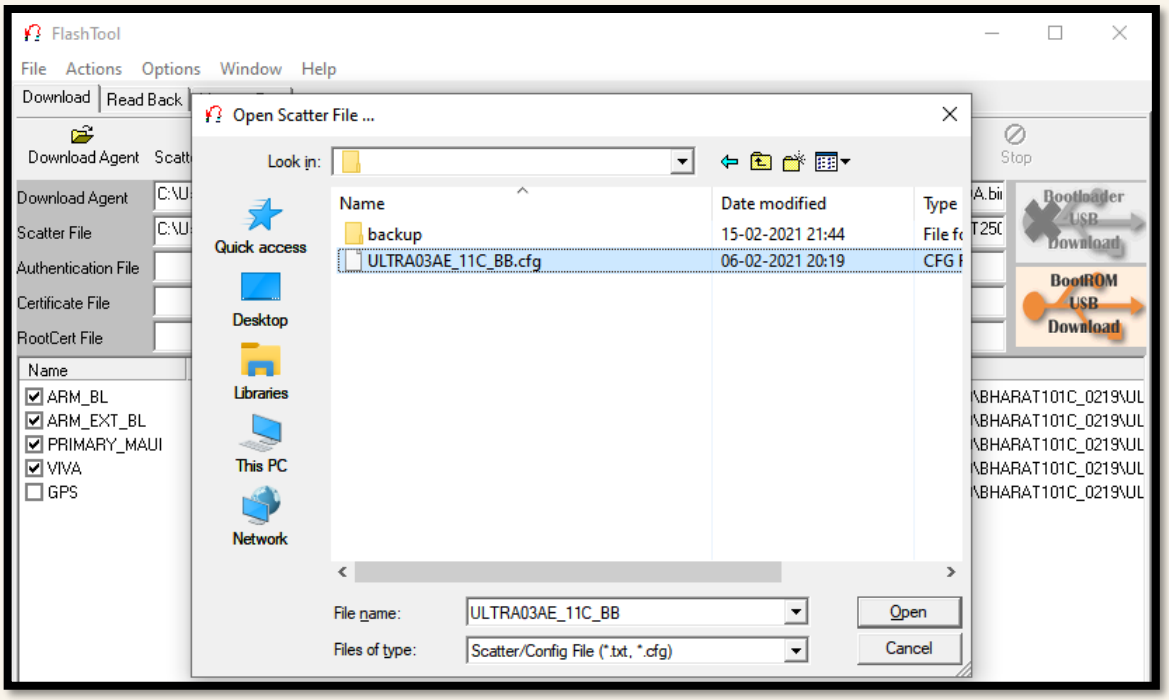
- To disable Format FAT,
  - Click the “Format FATxxxxxxx” option.
  - Tick the Do not Format FAT check box.
  - Untick Validation.
  - Click on the OK button as shown in the below images.



12.3.2.2 Loading Scatter File

The below process should be followed for loading scatter file.

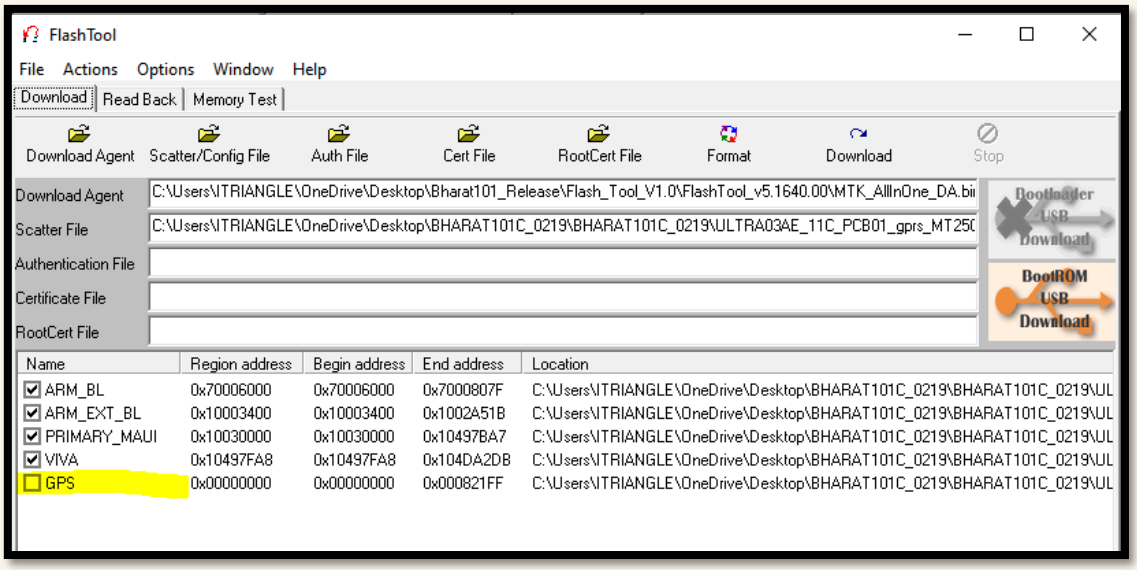
- Click on Button “Scatter/Config File”
- Select the .cfg with core whichever you want to download to the device from the firmware which is provided by FAE/ iTriangle support team.



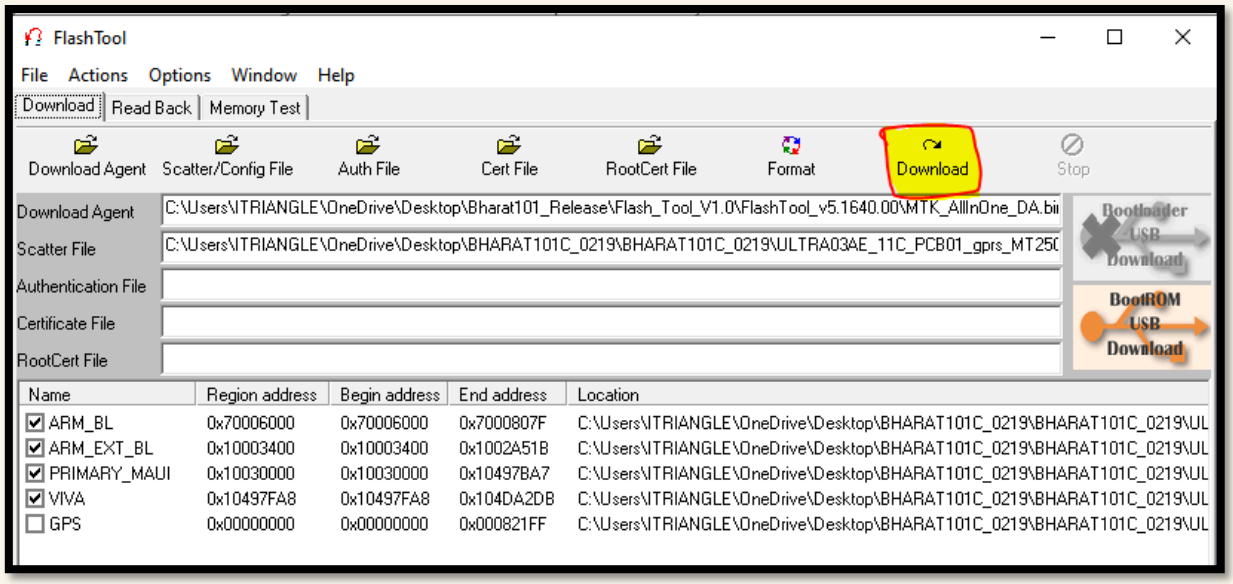
12.3.2.3 Download the Firmware

Once the scatter file is loaded follow the below process to download the firmware.

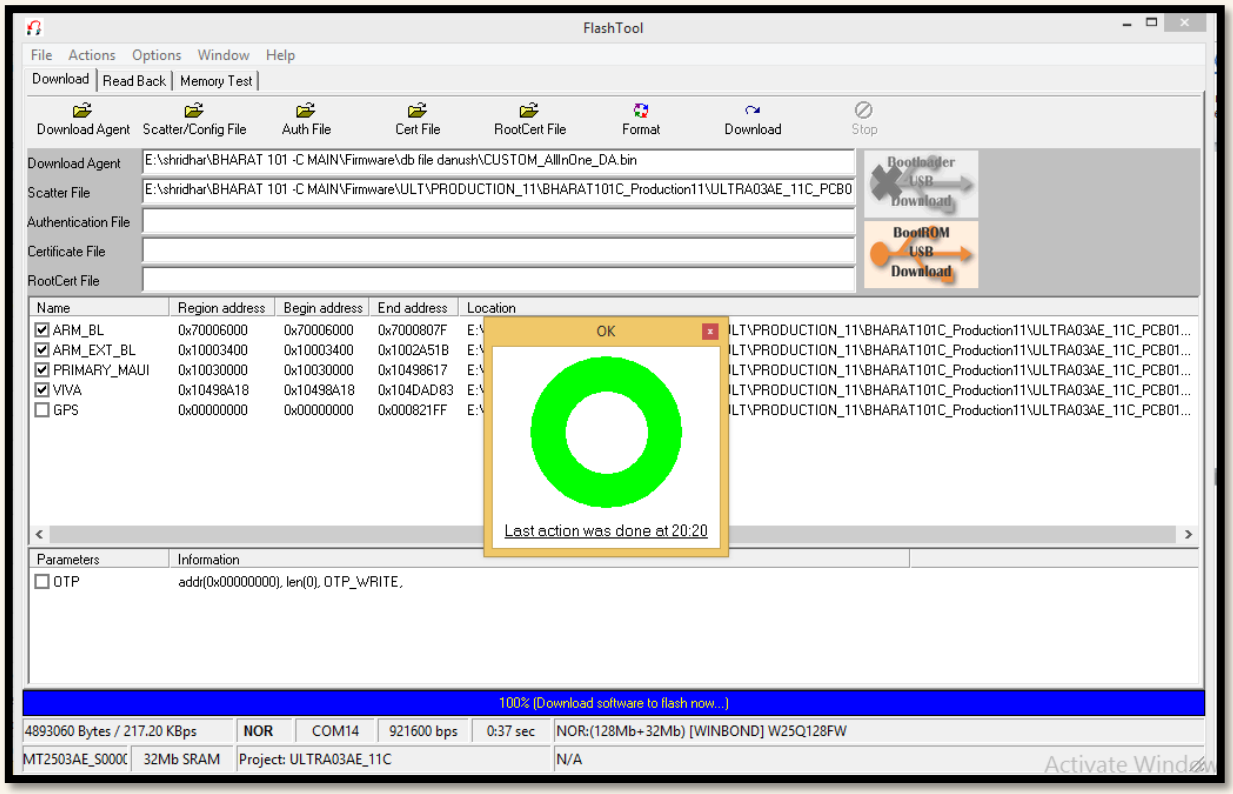
- Untick the GPS from the default files which are loaded as highlighted in the below image.



- Click on Download.



- Restart the device by pressing the RESET button. As soon as the reset button is pressed the Flash tool starts downloading the different files on to the device which will be shown in the status bar. After successful download of the firmware OK indication will be shown on the tool as shown below image.



- Restart the device again using the RESET button to start the device working on the latest firmware.
- You can verify the updated firmware version using the device configuration tool.

13 Settings and Configuration

The setting and configuration are supported from the below channels. For each command, individual responses will be sent from the device. The configuration response will be sent to the server if the configuration of the device is changed from any of the below sources along with the source name and source address if any. Other than USB for remaining sources, the address is mandatory.

The unit of all the interval/time configuration is in seconds unless it is specified. The device will not provide any response to the command if the format of the command which is received at the device is wrong.

It is preferred to use config 360 for the configuration of the device. The command version of the configuration should be used only in the case of SMS configuration.

Kindly make sure that you will verify all the configurations before the device is installed in the vehicle. iTriangle will not take responsibility for the parameters/values which are configured in the device.

13.1 Command Format

13.1.1 set

Set commands are used to configure the configuration parameters. Commands with both single and multiple keys should be executed and responded to the source of the command.

13.1.1.1 Single key

The Single key will be used to configure single parameters which will be having only one key and its value.

13.1.1.2 Syntax

$<random_number>,<IMEI>,<PASSWORD>,set,<command_key>:<value>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- Password: Password of the device.
- set: Indication of operation type.
- command_key: command identifier.
- Value: Value of the command which is under configuration.

As per the AIS-140 standard, the parameter change information and command execution status should be sent in the packet with predefined event code. A separate packet should be generated for each key and sent to the server. This logic is applicable irrespective of any source.

13.1.1.3 Response Format

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,set,<command_key>:< Command staus>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header Error if any.
- Command_source: the source of the command.
- Source_ID: Source ID
- set: Indicator of a set operation.
- command_key: command identifier.
- Command_status: Status of the command.
  - 1: Success.
  - -1: Failure due to the wrong key.
  - -2: Failure due to value out of range.
- Checksum: 8 bit XOR checksum.

Example:

1. **Success Case**
   Command:
   $123,860906040263136,aquila123,set,130:5*

   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,130:1*2B

2. **Failure due to the wrong key**
   Command:
   $123,860906040263136,aquila123,set,13:5*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,13:-1*36

3. Failure due to value out of range.
   Command:
   $123,860906040263136,aquila123,set,130:0*

   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,130:-2*05

13.1.1.4 Multiple Key

The multiple key formats will be used to configure the multiple parameters in a single command. By using multiple key format maximum of 15 keys can be executed by using a single command.

13.1.1.4.1 SYNTAX

$<random_number>,<IMEI>,<PASSWORD>,set,<command_key1>:<value1>,<command_key2:<value2>,<command_key3>:<value3>,.....,<command_key15>:<value15>*

Syntax parameter is as follows:

- random_number: Reference number for the command

- IMEI: IMEI/serial number of the device.
- password: Password of the device.
- set: Indicator of the set operation.
- command_key: command identifier.
- valueX: the value of the command which is under configuration.

Response Format

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,set,<command_key1>:<command_status1>,<command_key2>:<command_status2>,<command_key3>:<command_status3>,.....,<command_key15>:<command_status15>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header error if any(As per section 13.4.1.6).
- Command_source: the source of the command.
- Source_ID: Source ID
- set: Indicator of a set operation.
- command_keyX: command identifier.
- Command_status: Status of the command.
  - 1: Success.
  - -1:Failure due to the wrong key.
  - -2: Failure due to value out of range.
- Checksum: 8bit XOR checksum of the data.

Example:

1. Success Case

   Command:
   $123,860906040263136,aquila123,set,130:5,133:5*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,130:1,133:1*3D

2. Failure due to the wrong key
   Command:
   $123,860906040263136,aquila123,set,13:5,33:5*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,13:-1,33:-1*09

3. Failure due to value out of range.
   Command:

$123,860906040263136,aquila123,set,130:0,133:216001*
Command Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,130:-2,133:-2*3D

13.1.2 get

Get commands are used to query the value of configured parameters. Commands with both single and multiple keys will be executed and respond to the source of the command.

13.1.2.1 Single key

The Single key will be used to get single parameters which will be having only one key.

13.1.2.2 Syntax

$<random_number>,<IMEI>,<PASSWORD>,get,<command_key>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- Password: Password of the device.
- get: Indication of operation type.
- command_key: command identifier.

As per the AIS-140 standard, the parameter change information and command execution status should be sent in the packet with predefined event code. A separate packet should be generated for each key and sent to the server. This logic is applicable irrespective of any source.

13.1.2.3 Response Format

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,get,<command_key>:<value>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header Error if any.
- Command_source: the source of the command.
- Source_ID: Source ID
- get: Indicator of a set operation.
- command_key: command identifier.
- Value: Status/Value of the command.
  - -1: Failure due to the wrong key.
- Checksum: 8 bit XOR checksum.

Example:

4. Success Case
   Command:

$123,860906040263136,aquila123,get,130*

Command Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,get,130:5*3B

5. Failure due to the wrong key
   Command:
   $123,860906040263136,aquila123,get,13*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,get,13:-1*22

13.1.2.4 Multiple Key

The multiple key formats will be used to configure the multiple parameters in a single command. By using multiple key format maximum of 15 keys can be executed by using a single command.

13.1.2.4.1 SYNTAX

$<random_number>,<IMEI>,<PASSWORD>,get,<command_key1>:<value1>,<command_key2>,<command_key3>,.....,<command_key15>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- password: Password of the device.
- get: Indicator of the set operation.
- command_key: command identifier.

13.1.2.4.2 RESPONSE FORMAT

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,get,<command_key1>:<Value/status1>,<command_key2>:<Value/status2>,<command_key3>:<Value/status3>,.....,<command_key15>:<Value/status15>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header error if any(As per section 13.4.1.6).
- Command_source: the source of the command.
- Source_ID: Source ID
- get: Indicator of a set operation.
- command_keyX: command identifier.
- Command_status: Status of the command.
  - -1:Failure due to the wrong key.
- Checksum: 8bit XOR checksum of the data.

Example:

4. Success Case
   Command:
   $123,860906040263136,aquila123,get,130,133*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,get,130:5,133:5*29

5. Failure due to the wrong key
   Command:
   $123,860906040263136,aquila123,get,13,33*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,get,13:-1,33:-1*28

13.1.3 clr

clr command will be used to clear the configured value to default state.

13.1.3.1 Single key

The Single key will be used to clear the single parameters which will be having only one key.

13.1.3.2 Syntax

$<random_number>,<IMEI>,<PASSWORD>,clr,<command_key>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- Password: Password of the device.
- clr: Indication of operation type.
- command_key: command identifier.

As per the AIS-140 standard, the parameter change information and command execution status should be sent in the packet with predefined event code. A separate packet should be generated for each key and sent to the server. This logic is applicable irrespective of any source.

13.1.3.3 Response Format

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,clr,<command_key>:<status>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header Error if any.
- Command_source: the source of the command.
- Source_ID: Source ID
- clr: Indicator of a set operation.
- command_key: command identifier.

- Value: Status/Value of the command.
  - 1: Success
  - -1: Failure due to the wrong key.
- Checksum: 8 bit XOR checksum.

Example:

1. Success Case
   Command:
   $123,860906040263136,aquila123,clr,130*

   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,clr,130:1*34

2. Failure due to the wrong key
   Command:
   $123,860906040263136,aquila123,clr,13*
   Command Response:
   $$CLIENT_1ZF,860906040263136,30,123,0,1,1,clr,13:-1*19

13.1.3.4 Multiple Key

The multiple key formats will be used to clear the multiple parameters in a single command. By using multiple key format maximum of 15 keys can be executed by using a single command.

13.1.3.4.1 SYNTAX

$<random_number>,<IMEI>,<PASSWORD>,clr,<command_key1>:<value1>,<command_key2>,<command_key3>,.....,<command_key15>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- password: Password of the device.
- clr: Indicator of the set operation.
- command_key: command identifier.

13.1.3.4.2 RESPONSE FORMAT

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,clr,<command_key1>:<Value/status1>,<command_key2>:<Value/status2>,<command_key3>:<Value/status3>,.....,<command_key15>:<Value/status15>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device.
- random_number: Reference number for the command.
- header_error: header error if any(As per section 13.4.1.6).
- Command_source: the source of the command.
- Source_ID: Source ID

- clr: Indicator of a set operation.
- command_keyX: command identifier.
- Command_status: Status of the command.
  - 1: Success
  - -1:Failure due to the wrong key.
- Checksum: 8bit XOR checksum of the data.

Example:

1. Success Case
Command:
$123,860906040263136,aquila123,clr,130,133*
Command Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,clr,130:1,133:1*22
2. Failure due to the wrong key
Command:
$123,860906040263136,aquila123,clr,13,33*
Command Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,clr,13:-1,33:-1*21

13.1.4 debug

The debug commands are the commands which will not save the values after the restart. Only a single key is applicable for debug commands.

13.1.4.1 Syntax

$<random_number>,<IMEI>,<PASSWORD>,debug,<command_key>:<Command value>*

Syntax parameter is as follows:

- random_number: Reference number for the command
- IMEI: IMEI/serial number of the device.
- password: Password of the device.
- debug Indicator of the debug operation
- command_key: command identifier.
- Command Value: is either Enable/Disable(1/0) for debug command.

13.1.4.2 Response Format

$$<Cleint_ID>,<IMEI>,<Event_Code>,<random_number>,<header_error>,<command_source>,<source_ID>,debug,<command_key>:<command_status>*<Checksum>

Syntax parameter is as follows:

- Cleint_ID: header of the response.
- IMEI: IMEI/serial number of the device
- random_number: Reference number for the command.
- debug Indicator of a debug operation.
- header_error: header Error if any.
- Command_source: the source of the command.
- Source_ID: Source ID

- debug: Indicator of the debug operation
- command_key: command identifier.
- Command_status: Status of the command.
  - 1: Success.
  - -1:Failure due to the wrong key.
  - -2:Failure due to value out of range.
- Checksum: 8bit XOR checksum of the data.

Example:

1. Success Case
   Command:
   $123,358250330000513,aquila123,debug,102:0*

   Command Response:
   $$CLIENT_1ZF,358250330000513,30,123,0,1,1,debug,102:1*39
2. Failure due to the wrong key
   Command:
   $123,358250330000513,aquila123,debug,10:1*

   Command Response:
   $$CLIENT_1ZF,358250330000513,30,123,0,1,1,debug,10:-1*26
3. Failure due to value out of range.
   Command:
   $123,358250330000513,aquila123,debug,102:3*

   Command Response:
   $$CLIENT_1ZF,358250330000513,30,123,0,1,1,debug,102:-1*14

13.1.5 diag

The diag commands are the commands which will be applicable only from the SMS and serial for which the response will be sent on SMS as per the format defined. Only a single key execution is applicable for the diag command.

13.1.5.1 Syntax

$<random_number>,<IMEI>,<PASSWORD>,diag,<command_key>*

Syntax parameter is as follows:
random_number: Reference number for the command
IMEI: IMEI/serial number of the device.
password: Password of the device.
diag: Indicator of the diag operation
command_key: command identifier.

Example:
$123,358250330000513,aquila123,diag,100*

13.1.5.2 Response syntax

The response will be as per the response which is defined in the individual diagnostic command section.

13.1.6 Header Errors

Below are the header errors which will be sent in case of any errors in the header information. If any errors are present in the header the command will not be executed.

SL No	Header Error	Significance.
1	-1	Wrong format of random number
2	-2	IMEI number mismatch
3	-3	Password mismatch
4	-4	Operation type mismatch

13.1.7 Source List

Below codes concerning the Command source and the source ID Information.

SL No	Source	Command Source	Source ID
1	USB	1	1
2	Mobile Number 1	2	1
3	Mobile Number 2	2	2
4	Mobile Number 3	2	3
5	Mobile Number 4	2	4
6	Mobile Number 5	2	5
7	Mobile Number 6	2	6
8	Mobile Number 7	2	7

9	Mobile Number 8	2	8
10	Mobile Number 9	2	9
11	Mobile Number 10	2	10
12	Non Whitelisted No	2	Mobile Number
13	Primary IP	3	1
14	Secondary IP	3	2
15	Maintenance IP	3	3
16	Bluetooth	4	1
17	Emergency IP	3	4
18	Control Centre Number	2	11

14 Config 360

Config 360 is the tool that will be used for the configuration of the device using multiple channels.

- Launch the Tool
- Login using User Credentials.
- Selection of Configuration Channel.
- Select the local port.
- Select Device Model.
- Enter the Device Password.
- Do the Required Operations.
- Write to flash.
- Restart the Device.

It is mandatory to restart the device whenever any configuration of the device is changed.

Overview:

The Below Fig1.0 shows the overview of the Config360 tool. in this window, the user has various option listed below

- **The firmware version:** This will show the Firmware version present in the device.
- **IMEI Number:** The IMEI number present in the device.

- **Change password:** This option is provided to change the device password as per user requirement, once the password is changed user has to use a new password to communicate/configure any parameter to the device.
- **Read from flash:** This option is used to read the current configuration information from the device.
- **Read from file:** This option is provided to load the configuration file which is already created using the write to File option shown in Fig2.0. Cross verify all the values of the parameters which are read before writing it to the flash.
- **Restart Device:** This option is provided to restart the device when restart is necessary. The default state of the restart option is “OFF” when the user clicks on the curser device will restart after the configuration is successful using the write to flash option.
- **Select all commands:** This option is provided to select all the commands in a single click for the configuration.
- **Deselect all commands:** This option is provided to deselect all the commands.
- **Continue:** Using this option user can go to parameter edit options as shown in Fig2.0.

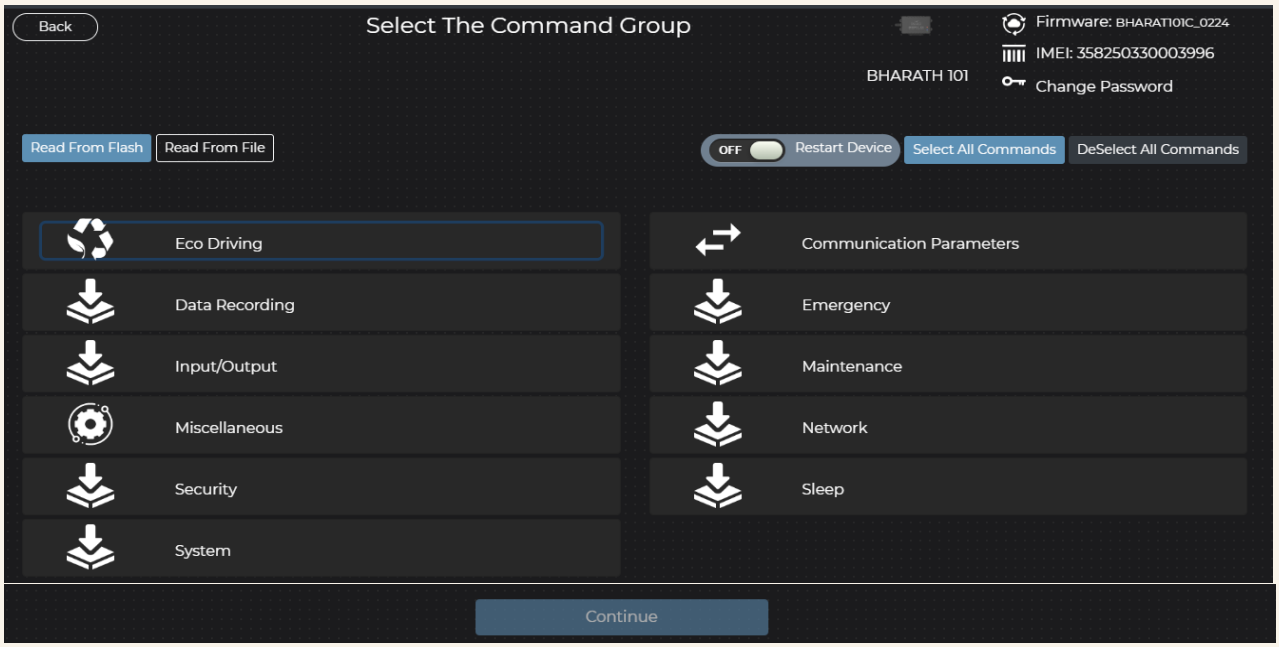


Fig 1.0

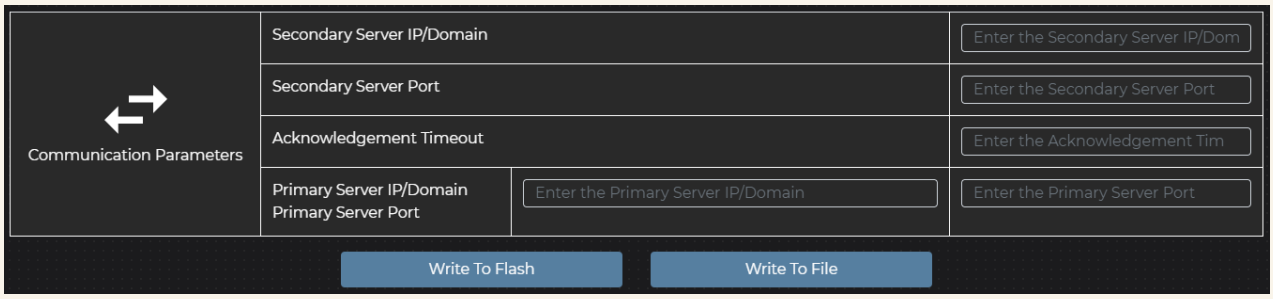


Fig 2.0

In Fig 2.0 user has two options.

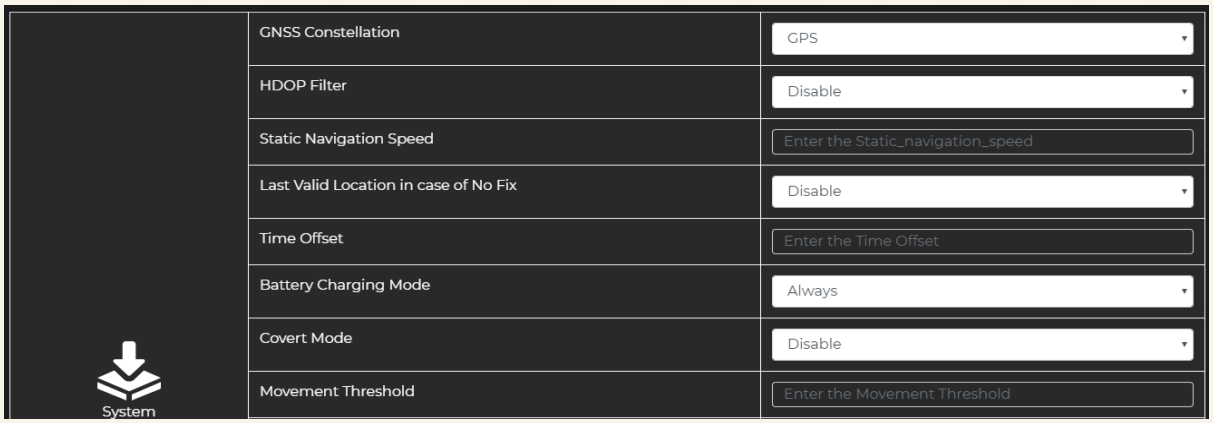
- **Write to Flash:** After the user editing the user configure parameter, the user can directly flash to the device by clicking on the write to Flash option. Kindly make sure that you will cross verify the values that you have configured.
- **Write to File:** After editing the user parameter user can create a configure parameter file by clicking on the write to File option. This is helpful when the user wants to configure the same parameters to many devices.

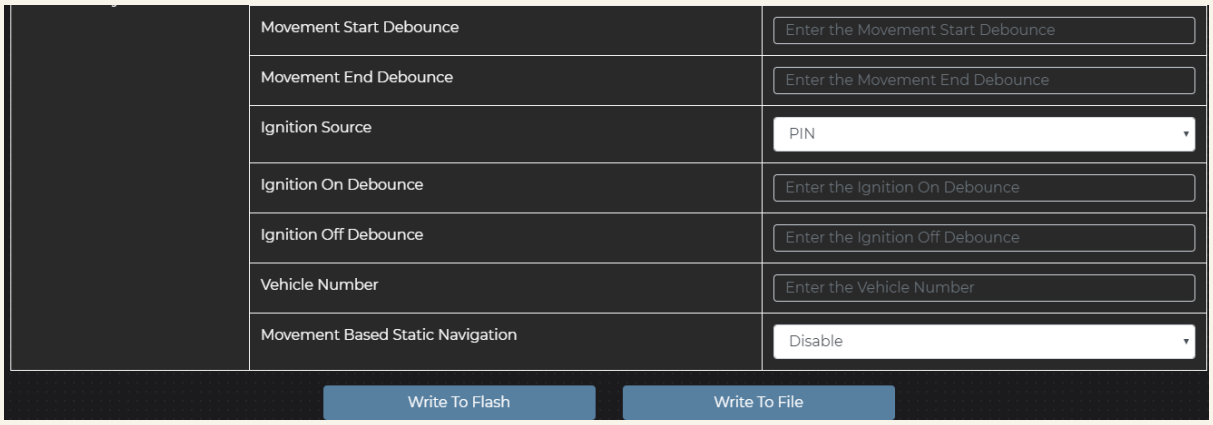
14.1 System

This section of the configuration contains the system-level configuration based on which other functionalities are dependant.

14.1.1 GNSS

In this section, all the GNSS/location update related commands explained in detail.





14.1.1.1 GNSS Constellation

The user can configure the device to use the different GNSS constellation for acquiring the location information. The device can use the below options for acquiring the location information.

- GPS
- GPS+IRNSS

14.1.1.1.1 USER COMMAND

1.1.1.1.1 SYNTAX:

$<Random_code>,<IMEI_Number>,<password>,set,147:<3/4>*

3:GPS constellation
4:(GPS+IRNSS) Constellations

Example:
$121,358250330000026,aquila123,set,147:4*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,147:1*2B

14.1.1.2 HDOP Filter

User can configure the device to enable or disable the HDOP filter along the HDOP filters threshold. When the HDOP value from the GNSS module crosses the configured value, the GNSS fix gets invalidated and the same information will be sent to the server. Only if the HDOP filter is enabled the threshold value which will be configured will applicable. There are 2 commands which are associated with it.

14.1.1.2.1 USER COMMAND

This command is used to enable or disable the HDOP Filter

Syntax:

$<Random_code>,<IMEI_Number>,<password>,set,2000:<0/1>*

0: Disable
1: Enable

Example:

$121,358250330000026,aquila123,set,2000:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2000:1*1B

14.1.1.3 HDOP Filter Value

This command is used to configure the HDOP value to the GNSS module. if we want to configure this HDOP value before we have to enable the HDOP to enable command(key:2000).

14.1.1.3.1 USER COMMAND

This command is used to configure the HDOP value. We can set this value from 1 to 99.

1.1.1.1.2 SYNTAX:

$<Random_code>,<IMEI_Number>,<password>,set,101:<10>*

Example:
$121,358250330000026,aquila123,set,101:10*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,101:1*29

14.1.1.4 Static Navigation Speed

Speed based static navigation is the functionality in which updated GNSS location information will be sent to the server only if the vehicle crosses the configured speed. This functionality will avoid the static wandering due to GNSS when the vehicle is in a static position. Below is the command to configure the static navigation speed.

14.1.1.4.1 USER COMMAND

This command is used to configure the Static Navigation Speed

1.1.1.1.3 SYNTAX:

$<Random_code>,<IMEI_Number>,<password>,set,102:2*

Example:
$121,358250330000026,aquila123,set,102:2*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,102:1*2A

14.1.1.5 Movement Based Static Navigation

Movement-based static navigation is the functionality in which the location of the vehicle is updated only in case if the movement is detected. If movement is not detected the latitude and longitude will not be updated in the packets. An option has been provided to enable/disable this functionality.

14.1.1.5.1 USER COMMAND

This command is used to enable/disable the Movement based static navigation.

Syntax:

$<Random_code>,<IMEI_Number>,<password>,set,2001:1/0*

0: Disable
1: Enable
Example:
$121,358250330000026,aquila123,set,2001:1*

Example:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2001:1*1A

14.1.1.6 Last Valid Location in case of No Fix

If this configuration is enabled, in case of no GNSS fix, the last valid latitude and longitude will be updated in the packets.

14.1.1.6.1 USER COMMAND

This command is used to enable/disable Last Valid Location in case of No Fix.

Syntax:

$<Random_code>,<IMEI_Number>,<password>,set,2002:1/0*

0: Disable
1: Enable
Example:
$121,358250330000026,aquila123,set,2002:1*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,2002:1*19

14.1.2 Time Offset

This command is used to configure the time offset against UTC that should be added in all the packets which are sent from the device.

14.1.2.1.1 USER COMMAND

This command is used to set the Time offset value in seconds.

Example:

$121,358250330000026,aquila123,set,110:19800*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,110:1*29

14.1.3 Battery Charging mode

The device is provided with a 3.7V internal Battery. This battery will provide backup power to the device whenever the main voltage supply is not available. The device has the option to configure the charging mode. There are charging modes that can be configured. One is always in which the battery charging will be always on. Another one is only during the ignition is ON.

14.1.3.1.1 USER COMMAND

This command is used to configure the Battery Charging mode.

Syntax:
$<Random_code>,<IMEI_Number>,<password>,set,119: 1/0*
0:Always
1:Only During Ignition On

Example:
$123,860906040263136,aquila123,set,119:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,119:1*20

14.1.4 Covert mode

The device has the option to configure the covert mode which will be helpful to covert the device and to minimise the current consumption of the device. If the covert mode is enabled all LEDs should be in OFF condition, otherwise LEDs should behave normally.

14.1.4.1 User command

This command is used to configure the Covert mode.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,2009:< 1/0>*
1: Covert Mode Enable
0: Covert Mode Disable

Example:
$123,860906040263136,aquila123,set,2009:0/1*

Response
$$CLIENT_1ZF, 358250330000026,30,121,0,1,1,set,2009:1*12

14.1.5 Movement Detection

The vehicle movement status can be identified by using the accelerometer which is embedded in the device. When the vibration is detected, for the configured debounce time, the device can consider it as the movement is detected in the vehicle. The engine is also considered as the movement if the configured threshold is crossed.

14.1.5.1 Movement Threshold

This command is used to configure the movement threshold is the device.

14.1.5.1.1 USER COMMAND.

This command is used to configure the Movement threshold.

Command:

$121,358250330000026,aquila123,set,143:20*

Response:

$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,143:1*2F

14.1.5.2 Movement Start Debounce

This command is used to configure the debounce interval for which the vibration should be detected to consider as a movement.

14.1.5.2.1 USER COMMAND

This command is used to configure the Movement to start to debounce.

Example:

$121,358250330000026,aquila123,set,144:1*

Response:

$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,144:1*28

14.1.5.3 Movement End Debounce

This command is used to configure the debounce interval for which the vibration should not be detected to consider as no movement.

14.1.5.3.1 USER COMMAND

This command is used to configure the Movement end debounce.

Example:

$121,358250330000026,aquila123,set,145:5*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,145:1*29

14.1.6 Ignition Detection

The device supports multiple sources that can be used to detect the ignition. Below are the different sources that can be used to detect the ignition.

- Ignition PIN
- Accelerometer
- Main power voltage
- Speed
- All Above

If the ignition detection source is configured as PIN, the dedicated ignition pin will be used as a source. If high is detected on this PIN the ignition status will be updated as ON and if the low is detected, the status will be updated as Off.

If the ignition detection source is configured as an accelerometer, the movement status will be used to update the ignition status. If the movement is detected, the status will be updated as ON and if the movement is not detected, the status will be updated as ofF.

If the ignition detection source is configured as the main power voltage, the main power voltage level will be used to update the ignition status. If the main power voltage is more than

configured threshold voltage, the status will be updated as ON and if the main power voltage is less than the configured threshold, the status will be updated as of.

If the ignition detection source is configured as speed, the speed received from GNSS will be used to update the ignition status. If the speed is more than configured threshold, the status will be updated as ON and if the speed is less than the configured threshold, the status will be updated as of.

If the ignition detection source is configured as all, all the above sources will be used for ignition detection. If any of the above sources detects the ignition, the status will be updated as ON and if all the above sources detect the ignition off, the status will be updated as of.

14.1.6.1 User Commands

Below are the commands which are used for Ignition detection.

14.1.6.1.1 SOURCE IGNITION

This command is used to configure the Ignition source.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,167:<0/1/2/3/4>*
0: PIN
1: Accelerometer
2: Main Power Voltage
3: Speed
4: All

Example:
$123,860906040263136,aquila123,set,167:0*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,167:1*29

14.1.6.1.2 IGNITION ON DEBOUNCE

This command is used to configure the debounce for detection of the ignition ON status. If the source should maintain the ON condition continuously for configured debounce interval, the ignition status will be detected as ON. Kindly make sure that you will choose the debounce wisely as distance calculation depends on ignition ON status.

Example:
$123,860906040263136,aquila123,set,169:1*

Example:
S$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,169:1*27

14.1.6.1.3 IGNITION OFF DEBOUNCE

This command is used to configure the debounce for detection of the ignition Off status. If the source should maintain the Off condition continuously for configured debounce interval, the ignition status will be detected as OFF.

Example:
$123,860906040263136,aquila123,set,170:1*

Example:
S$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,170:1*2F

14.1.6.1.4 VOLTAGE THRESHOLD

This command is used to configure the Voltage Threshold for voltage based ignition detection.

Example:
$123,860906040263136,aquila123,set,168:12000*

Example:
S$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,168:1*26

14.1.6.1.5 SPEED THRESHOLD

This command is used to configure the Speed Threshold for speed-based ignition detection.

Example:
$123,860906040263136,aquila123,set,171:5*

Example:
S$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,171:1*2E

14.1.7 Vehicle Number

The device has the option to configure the vehicle number. The configured vehicle number will be reflected in the respective field in the packets.

14.1.7.1 User command

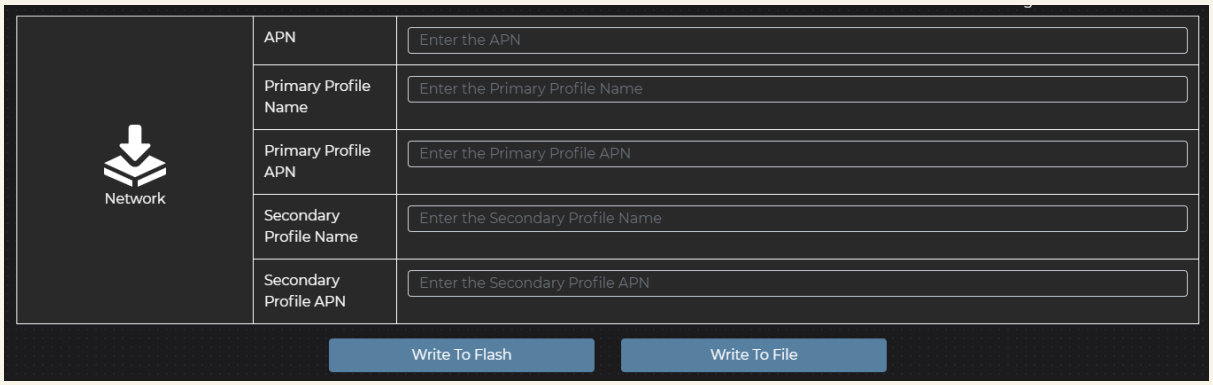
This command is used to configure the vehicle number.

Example:
$123,860906040263136,aquila123,set,211:KA04KB1234*

Response:
S$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,211:1*2B

14.2 Network

This section explains the network-related configuration which is available in the device.



14.2.1 APN

This command is used to configure the common APN. This APN will be used to activate the Bearer connection with the network. This APN will be used only if a profile based APN is not available for specific profiles.

14.2.1.1 User Command

This command is used to configure the APN.

Example:

$123,358250330000513,aquila123,set,103:taisysnet*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,103:1*2B

14.2.2 Primary Profile Name

The Primary profile can configurable with the sim profile name. This profile is used to identify the current profile of the network and select the respective APN to use for bearer activation.

14.2.2.1 User Command

This command is used to configure the primary profile name. Kindly make sure that the primary profile name is configured as per the primary SIM profile name. This configuration will be used for

SIM switching and usage of the APN. The wrong configuration might lead to non-communication of the device.

Example:
$123,358250330000513,aquila123,set,191:airtel *

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,191:1*20

14.2.3 Primary profile APN

The primary profile APN is used to configure the APN of the primary profile. The device will use the primary profile APN if the current profile is identified as the primary profile.

14.2.3.1 User Command

This command is used to configure the primary profile APN.

Example:
$123,358250330000513,aquila123,set,192:taisysnet*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,192:1*23

14.2.4 Secondary profile name

The secondary profile can configurable with the sim profile name. Kindly make sure that the primary profile name is configured as per the secondary SIM profile name. This configuration will be used for SIM switching and usage of the APN. The wrong configuration might lead to non-communication of the device.

14.2.4.1 User Command

This command is used to configure the Secondary profile name.

Example:

$123,358250330000513,aquila123,set,193:BSNL*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,193:1*22

14.2.5 Secondary profile APN

The secondary profile APN is used to configure the APN of the secondary profile. The device will use the secondary profile APN if the current profile is identified as the primary profile.

14.2.5.1 User Command

This command is used to configure the Secondary profile APN.

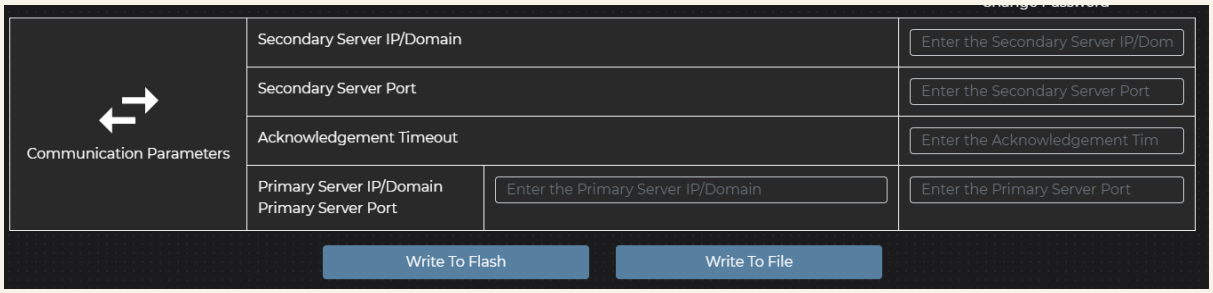
Example:

$123,358250330000513,aquila123,set,194:bsnlnet*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,set,194:1*25

14.3 Communication

This section explains the communication-related parameters which can be configured in the device. The device is capable of communication over GPRS, SMS. Over GPRS as per AIS-140 standard, the device can communicate to 3 servers simultaneously.



14.3.1 Primary server IP/Domain

This command is used to configure the IP/domain name of the primary service for which the device has to communicate.

14.3.1.1 User Command

Example:
$108,358250330000026,aquila123,set,111:dev.aquilatrack.com*

Response:
$$CLIENT_1ZF,358250330000026,30,108,0,1,1,set,111:1*28

14.3.2 Primary server Port

This command is used to configure the port of the primary server to which the device has to communicate.

14.3.2.1 User Command

Example:
$108,358250330000026,aquila123,set,112:12345*

Response:
$$CLIENT_1ZF,358250330000026,30,108,0,1,1,set,112:1*2B

14.3.3 Secondary server IP/Domain

This command is used to configure the IP/Domain of the secondary server to which the device has to communicate.

14.3.3.1 User Command

Example:
$108,358250330000026,aquila123,set,114:w.aquilatrack.com*

Response:
$$CLIENT_1ZF,358250330000026,30,108,0,1,1,set,114:1*2D

14.3.4 Secondary server Port

This command is used to configure the port of the primary server to which the device has to communicate.

14.3.4.1 User Command

Example:
$108,358250330000026,aquila123,set,115:12345*

Response:
$$CLIENT_1ZF,358250330000026,30,108,0,1,1,set,115:1*2C

14.3.5 Packet Acknowledgement Timeout

The device has the option to configure the socket acknowledgement timeout for the packet which is sent. After the packet is sent the device will wait for the timeout which is configured. If the acknowledgement is not received within configured timeout the packet will be a move to history and will send it as a history packet. This configuration should be done wisely as lowering the timeout might lead to duplicate packets.

14.3.5.1 User Command

Example:
$108,358250330000026,aquila123,set,116:60*

Response:
$$CLIENT_1ZF,358250330000026,30,108,0,1,1,set,116:1*2F

14.3.6 Common Layer

The Device has option to operate in different type of common layer functionality.

14.3.6.1 User Command

Below are the commands which are used for common layer selection.

14.3.6.1.1 COMMON LAYER

This command is used to configure the common layer.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,190:<0/1/2/5>*
0: Standard AIS140
1: APSAC
2: Uttarakhand
5: Rajasthan

Example:
$123,860906040263136,aquila123,set,290:0*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,290:1*32

14.4 Emergency

This functionality is used for emergency purposes. Once the panic button is pressed, the device will move to emergency mode and send an emergency alert to all three servers with all the necessary fields like latitude, longitude, their directions, and other information described in AIS140 compliance. Only emergency packets will be sent to the emergency server. If the connection to the emergency server is not available, the device will start sending the command

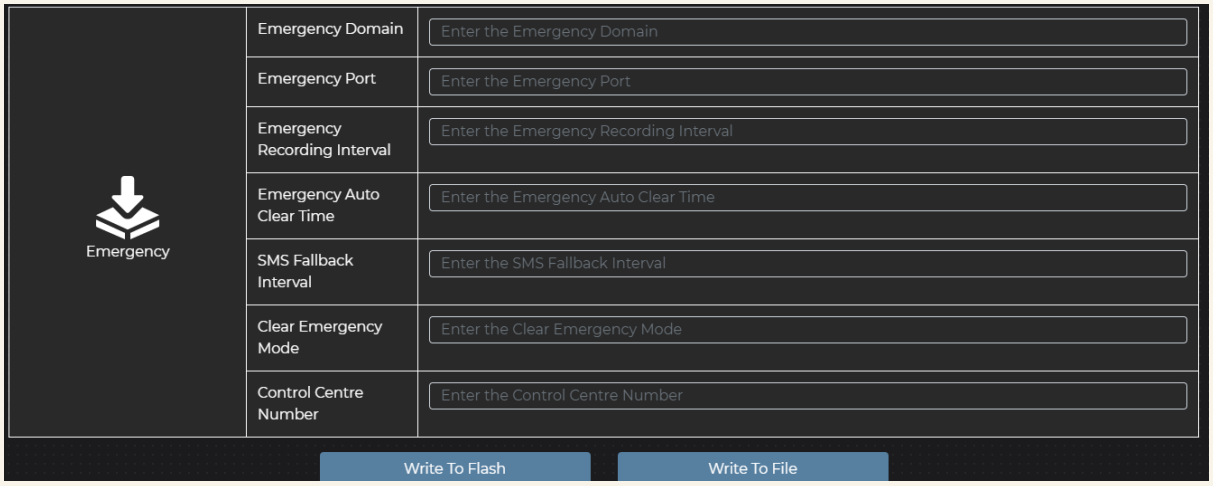
in SMS format to the control centre number as per the fall back SMS interval. When the device moves to emergency mode the tracking interval of the secondary server will also be changed to an emergency interval.

The emergency mode will be cleared in case if the emergency clear command is received from the emergency server or the emergency auto clear time is expired.

If the emergency wire is disconnected for 20 seconds, the tamper alert will get generated and all the tracking packets will be sending the event code 16.

14.4.1 Emergency server IP/Domain

This command is used to configure the IP/Domain of the secondary server to which the device has to communicate.



14.4.1.1 User command

Example:
$123,860906040263136,aquila123,set,195:106.51.85.249*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,195:1*24

14.4.2 Emergency server port

This command is used to configure the port of the primary server to which the device has to communicate.

14.4.2.1 User command

Example:
$123,860906040263136,aquila123,set,196:17003*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,196:1*27

14.4.3 Emergency Recording Interval

The emergency recording interval will be used to configure the interval of packet recording for the emergency server when the device is in emergency mode.

14.4.3.1 User command

Example:
$123,860906040263136,aquila123,set,207:10*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,207:1*2C

14.4.4 Emergency Auto Clear Time

This command is used to configure the automatic clear timeout to clear the emergency mode automatically.

14.4.4.1 User command

Example:
$123,860906040263136,aquila123,set,208:600*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,208:1*23

14.4.5 SMS Fallback Interval.

This command is used to configure the SMS fall back interval in case the emergency server is not connected when the device is in emergency mode.

14.4.5.1 User command

Example:
$123,860906040263136,aquila123,set,209:300*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,209:1*22

14.4.6 Control Centre Number

This command is used to configure the control centre number which will be used to send emergency SMS fallback SMS when the emergency server is not connected.

14.4.6.1 User command

Example:
$123,860906040263136,aquila123,set,236:+919900000000*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,236:1*2E

14.4.7 Emergency Clear

This command is used to clear the emergency mode. This command will be executed either from the control centre number or the emergency server.

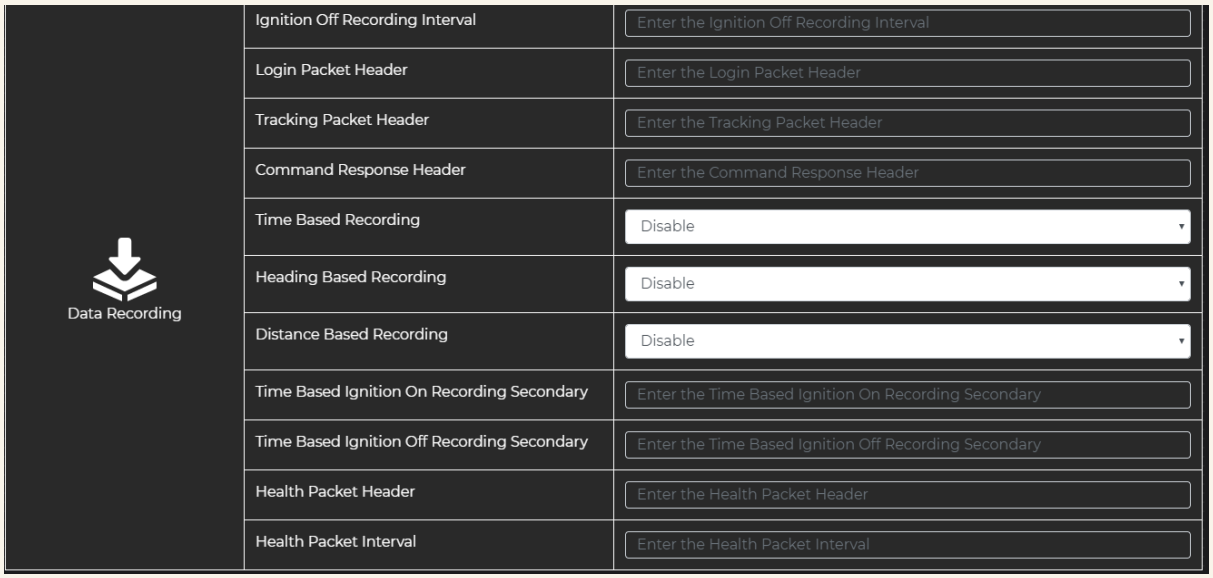
14.4.7.1 User command

Example:
$123,860906040263136,aquila123,set,2040:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2040:1*1F

14.5 Data Recording

The device is having an option to record the data based on different modes. The data recording can be done based on the time interval, heading based and distance-based.



14.5.1 Time-based recording

The device will start recording packets as per the interval which is configured. There is an option to configure different time-based intervals for the primary and secondary server.

14.5.1.1.1 IGNITION ON INTERVAL FOR PRIMARY

This command is used to set the ignition on the interval for the Primary server.

Example:
$121,358250330000026,aquila123,set,130:5*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,130:1*2B

14.5.2 Ignition Off Interval for Primary

This command is used to set the ignition OFF interval for the Primary server.

Example:
$121,358250330000026,aquila123,set,133:60*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,133:1*28

14.5.3 Ignition On Interval for Secondary

This command is used to set the ignition on the interval for the secondary server.

Example:
$121,358250330000026,aquila123,set,205:5*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,205:1*2E

14.5.4 Ignition Off Interval for Secondary

This command is used to set the ignition off interval for a secondary server.

Example:
$121,358250330000026,aquila123,set,206:60*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,206:1*2D

14.5.5 Heading Based Recording

The device has the option to record the packet when the change in heading crosses the threshold which is configured. The device will record the heading based packet when heading based recording is enabled, the vehicle is in motion and the speed should be greater than 10kmph. Heading based recording is Disable, the device will record the data packet on the configured time interval. This functionality is only applicable to the primary server.

14.5.5.1 User Commands

14.5.5.1.1 ENABLE/DISABLE

This command is used to Enable/Disable the Heading base Recording for the primary server.

Syntax:
$<Random_code>,<IMEI_Number>,<password>,set,2023:1/0*
0: Disable
1: Enable

Example:
$121,358250330000026,aquila123,set,2023:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2023:1*1A

14.5.5.1.2 THRESHOLD

This command is used to configure the threshold for heading based recording.

Example:
$121,358250330000026,aquila123,set,2023:1,132:100*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2023:1,132:1*29

14.5.6 Distance-Based Recording

The device has the option to record the packet when the vehicle travels configured distance from the latest distance-based recorded packet. The device will record the distance-based packet when distance-based recording is enabled and the distance accumulated is more than the configured threshold. If the distance-based recording is Disabled, the device will record the data packet on the configured time interval. This functionality is only applicable to the primary server.

14.5.6.1 User Commands

14.5.6.1.1 ENABLE/DISABLE

This command is used to Enable/Disable the Distance base Recording for the primary server.

Syntax:
$<Random_code>,<IMEI_Number>,<password>,set,2024:1/0*
0: Disable
1: Enable

Example:
$121,358250330000026,aquila123,set,2024:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2024:1*1D

14.5.6.1.2 DISTANCE THRESHOLD

This command is used for configuring the Distance in the meter.

Example:
$121,358250330000026,aquila123,set,2024:1,132:100*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,132:1*29

14.5.7 Login packet Header

The device has the option to configure the header of the login packet. This command is used to configure the header as per user requirement. The header which is configured will be the same for both primary and secondary servers.

14.5.7.1 User command

This command is used to configure the header of the login packet.

Example:
$123,860906040263136,aquila123,set,163:Header*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,163:1*2D

14.5.8 Tracking Packet Header

The device has the option to configure the header of the tracking packet. This command is used to configure the header as per user requirement. The header which is configured will be the same for both primary and secondary servers.

14.5.8.1 User command

This command is used to configure the header of the tracking packet.

Example:
$123,860906040263136,aquila123,set,164:Header*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,164:1*2A

14.5.9 Health Packet

The health packet will be recorded as per the interval configured by the user. The device has the option to configure the header of the health packet. This command is used to configure the header as per user requirement. The header which is configured will be the same for both primary and secondary servers.

14.5.9.1 User Commands

14.5.9.1.1 HEADER

This command is used to configure the Health Packet Header.

Example:
$123,860906040263136,aquila123,set,243:Header*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,243:1*2C

14.5.9.1.2 INTERVAL

This command is used to configure the Health packet interval for both the primary and secondary server.

Example:
$123,860906040263136,aquila123,set,244:300*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,244:1*2B

14.5.10 Command Response Header

The device has the option to configure the header of the command response packet. This command is used to configure the header as per user requirement. The header which is configured will be the same for both primary and secondary servers.

14.5.10.1 User command

This command is used to configure the header of the Command Response.

Example:
$123,860906040263136,aquila123,set,165:CLIENT_1ZF*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,165:1*2B

14.5.11 Datatype Change (Heading and Odometer)

The device has the option to configure the data type of the field Heading and the odometer. This is used to keep the backward compatibility of the protocol.

14.5.11.1 User Command

This command is used to configure the data type of the Heading and the odometer field in the packet protocol.

Example:
$123,860906040263136,aquila123,set,2064:<0/1>*
Where:
0: Integer
1: Float

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2064:1*2B

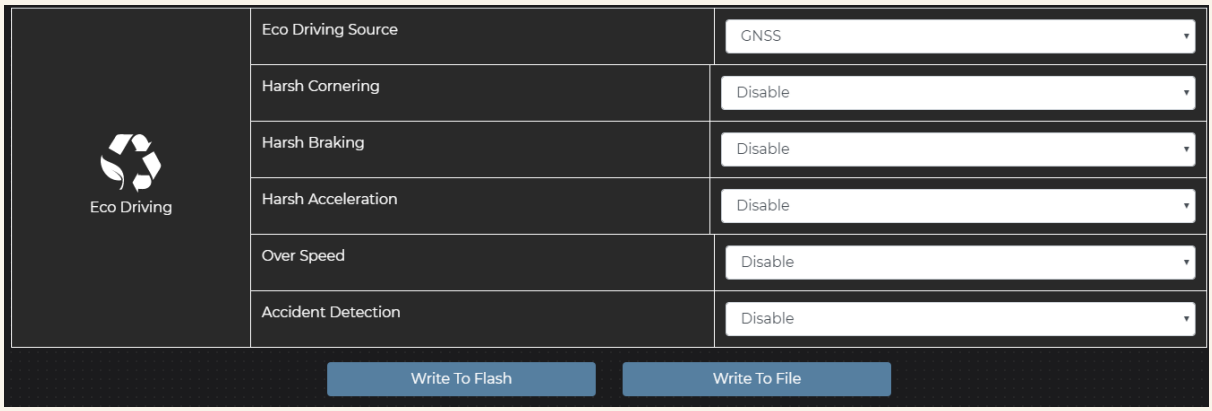
14.6 Eco Driving

The eco-driving in simple words is ecological, economical and safe driving. The device gives the facility to detect the driving behaviour of the driver. The eco-driving includes the below events which get generated based on different configurations.

- Harsh Acceleration
- Harsh Braking
- Harsh Cornering
- Overspeed
- Accident Detection

The device enables the user to configure the eco-driving source. The option of source selection will be applicable only for a generation of the HA, HB and HC. The rest of the events are not dependant on the eco-driving source. 2 sources can be used for eco-driving.

- Accelerometer
- Speed



14.6.1 Eco Driving Source

This command is used to configure the ECO driving Source.
Command Syntax:
$<random_number>,<IMEI>,<password>,set,<2014>:1/0*
1: Accelerometer
0: GNSS Based

Example:
$$123,860906040263136,aquila123,set,2014:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2014:1*1E

14.6.2 Eco Driving Axis

This command is used to configure the Eco Driving Axis.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,<285>:0/1/2*
0: X-Axis
1: Y-Axis
2: Z-Axis
Example:
$$123,860906040263136,aquila123,set,285:0*
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,285:1*26

14.6.3 Harsh Acceleration

Harsh acceleration event will be detected only if the Harsh Acceleration is enabled. Harsh acceleration will be generated when the vehicle accelerates more than the configured threshold of event generation. Below are the configurations which are applicable for the generation of Harsh Acceleration.

14.6.3.1 User command

14.6.3.1.1 ENABLE/DISABLE

This command is used to enable/disable the generation of harsh acceleration event. This command is applicable independent of the eco-driving source.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,<2011> :<1/0>*
0:Disable
1:Enable

Example
$123,860906040263136,aquila123,set,2011:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2011:1*1B

14.6.3.1.2 THRESHOLD

This command is used to configure the threshold for the harsh acceleration. This configuration is independent of the eco-driving source. The threshold should be configured with the unit rate of change of speed per second. Which is kmph/sec.

Example:
$123,860906040263136,aquila123,set,190:10*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,190:1*21

14.6.3.1.3 DEBOUNCE

The debounce for harsh acceleration is dependent on the eco-driving source.

14.6.3.1.3.1 ACCELEROMETER DEBOUNCE

This command is used to configure the debounce when the eco-driving source is configured as an accelerometer-based harsh acceleration debounce. The accelerometer-based debounce configuration will be done in seconds with decimal points.

Example:
$123,860906040263136,aquila123,set,260:0.5*
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,260:1*2D

14.6.3.1.3.2 GNSS DEBOUNCE

This command is used to configure the harsh acceleration debounce when the eco-driving source is configured as GNSS. The GNSS based debounce configuration will be done in seconds.

Example:
$123,860906040263136,aquila123,set,263:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,263:1*2E

14.6.3.1.4 HA INDICATION ON DOUT1

Once Harsh Acceleration event is generated, the indication can be provided on the DOUT1. To enable this the DOUT1 functionality should be configured as Eco Driving Indicator. The indication can be provided with different patterns.

- Enable for 5 Sec
- Enable with 500ms ON time with 50% Duty Cycle
- Enable with 1sec ON time with 50% Duty Cycle
- Enable with 2 sec ON time with 50% Duty cycle
- Enable with 3 sec ON time with 50% Duty Cycle

Example:
$123,860906040263136,aquila123,set,152:<0/1/2/3/4/5>*
Where:
0: Disable
1: Enable for 5 Sec
2: Enable with 500ms ON time with 50% Duty Cycle
3: Enable with 1sec ON time with 50% Duty Cycle
4: Enable with 2 sec ON time with 50% Duty cycle
5: Enable with 3 sec ON time with 50% Duty Cycle

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,152:1*2E

14.6.4 Harsh Braking

The harsh braking event will be detected only if the Harsh Barking is enabled. Harsh braking will be generated when the vehicle decelerates more than the configured threshold of event generation. Below are the configurations which are applicable for the generation of Harsh Braking.

14.6.4.1 User command

14.6.4.1.1 ENABLE/DISABLE

This command is used to enable/disable the generation of a harsh braking event. This command is applicable independent of the eco-driving source.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,2012:<1/0>*
0:Disable
1:Enable

Example
$123,860906040263136,aquila123,set,2012:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2012:1*18

14.6.4.1.2 THRESHOLD

This command is used to configure the threshold for harsh braking. This configuration is independent of the eco-driving source. The threshold should be configured with the unit rate of change of speed per second. Which is kmph/sec.

Example:
$123,860906040263136,aquila123,set,148:10*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,148:1*24

14.6.4.1.3 DEBOUNCE

The debounce for harsh braking is dependant on the eco-driving source.

14.6.4.1.3.1 ACCELEROMETER DEBOUNCE

This command is used to configure the debounce when the eco-driving source is configured as an accelerometer-based harsh braking debounce. The accelerometer-based debounce configuration will be done in seconds with decimal points.

Example:
$123,860906040263136,aquila123,set,261:0.5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,261:1*2C

14.6.4.1.3.2 GNSS DEBOUNCE

This command is used to configure the harsh braking debounce when the eco-driving source is configured as GNSS. The GNSS based debounce configuration will be done in seconds.

Example:
$123,860906040263136,aquila123,set,264:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,264:1*29

14.6.4.1.4 HB INDICATION ON DOUT1

Once Harsh Braking event is generated, the indication can be provided on the DOUT1. To enable this the DOUT1 functionality should be configured as Eco Driving Indicator. The indication can be provided with different patterns.

- Enable for 5 Sec
- Enable with 500ms ON time with 50% Duty Cycle
- Enable with 1sec ON time with 50% Duty Cycle
- Enable with 2 sec ON time with 50% Duty cycle
- Enable with 3 sec ON time with 50% Duty Cycle

Example:
$123,860906040263136,aquila123,set,153:<0/1/2/3/4/5>*

Where:
0: Disable
1: Enable for 5 Sec
2: Enable with 500ms ON time with 50% Duty Cycle
3: Enable with 1sec ON time with 50% Duty Cycle
4: Enable with 2 sec ON time with 50% Duty cycle
5: Enable with 3 sec ON time with 50% Duty Cycle

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,153:1*2E

14.6.5 Harsh Cornering

Harsh cornering event will be detected only if the Harsh Cornering is enabled. Harsh cornering will be generated when the vehicle corners more than the configured threshold of event generation. Below are the configurations which are applicable for the generation of Harsh cornering.

14.6.5.1 User command

14.6.5.1.1 ENABLE/DISABLE

This command is used to enable/disable the generation of harsh cornering event. This command is applicable independent of the eco-driving source.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,2013:<1/0>*

0:Disable
1:Enable

Example
$123,860906040263136,aquila123,set,2013:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2013:1*19

14.6.5.1.2 THRESHOLD

This command is used to configure the threshold for the harsh cornering. This configuration is independent of the eco-driving source. The threshold should be configured with the unit rate of change of angle per second. Which is degree/sec.

Example:
$123,860906040263136,aquila123,set,149:10*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,149:1*25

14.6.5.1.3 DEBOUNCE

The debounce for harsh cornering is dependant on the eco-driving source.

14.6.5.1.3.1 ACCELEROMETER DEBOUNCE

This command is used to configure the debounce when the eco-driving source is configured as an accelerometer-based harsh cornering debounce. The accelerometer-based debounce configuration will be done in seconds with decimal points.

Example:
$123,860906040263136,aquila123,set,262:0.5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,262:1*21

14.6.5.1.3.2 GNSS DEBOUNCE

This command is used to configure the harsh cornering debounce when the eco-driving source is configured as GNSS. The GNSS based debounce configuration will be done in seconds.

Example:
$123,860906040263136,aquila123,set,265:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,265:1*2F

14.6.5.1.4 HC INDICATION ON DOUT1

Once Harsh Cornering event is generated, the indication can be provided on the DOUT1. To enable this the DOUT1 functionality should be configured as Eco Driving Indicator. The indication can be provided with different patterns.

- Enable for 5 Sec
- Enable with 500ms ON time with 50% Duty Cycle
- Enable with 1sec ON time with 50% Duty Cycle
- Enable with 2 sec ON time with 50% Duty cycle
- Enable with 3 sec ON time with 50% Duty Cycle

Example:
$123,860906040263136,aquila123,set,155:<0/1/2/3/4/5>*
Where:
0: Disable
1: Enable for 5 Sec
2: Enable with 500ms ON time with 50% Duty Cycle
3: Enable with 1sec ON time with 50% Duty Cycle
4: Enable with 2 sec ON time with 50% Duty cycle
5: Enable with 3 sec ON time with 50% Duty Cycle

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,155:1*2E

14.6.6 Over speed

This event will generate only when the current speed is more than the configured threshold continuously for debounce time.

14.6.6.1 User command

14.6.6.1.1 ENABLE/DISABLE

This command is used to enable/disable the Overspeed event generation by the device.

Command Syntax: $<random_number>,<IMEI>,<password>,set,2015:<1/0>*

0:Disable
1:Enable

Example:
$123,860906040263136,aquila123,set,2015:1*

Response:
$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2015:1*1F

14.6.6.1.2 DEBOUNCE

This command is used to configure the over speed debounce.

Example:
$123,860906040263136,aquila123,set,150:80*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,150:1*2D

14.6.6.1.3 USER COMMAND

This command is used to configure the Over Speed Debounce Time for an over-speed event.

Example:
$123,860906040263136,aquila123,set,151:5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,151:1*2C

14.6.6.1.4 OVERSPEED INDICATION ON DOUT1

Once overspeed event is generated, the indication can be provided on the DOUT1. To enable this the DOUT1 functionality should be configured as Eco Driving Indicator. The indication can be provided with different patterns.

- Enable for 5 Sec
- Enable with 500ms ON time with 50% Duty Cycle
- Enable with 1sec ON time with 50% Duty Cycle
- Enable with 2 sec ON time with 50% Duty cycle
- Enable with 3 sec ON time with 50% Duty Cycle

Example:
$123,860906040263136,aquila123,set,156:<0/1/2/3/4/5>*

Where:
0: Disable
1: Enable for 5 Sec
2: Enable with 500ms ON time with 50% Duty Cycle
3: Enable with 1sec ON time with 50% Duty Cycle
4: Enable with 2 sec ON time with 50% Duty cycle
5: Enable with 3 sec ON time with 50% Duty Cycle

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,156:1*2E

14.6.7 Accident Detection

The device is provided with accident detection functionality. this event will detect when the impact is observed in any directions.

14.6.7.1 User command

14.6.7.1.1 ENABLE/DISABLE

This command is used to Enable/disable accident detection.

Syntax:
$<random_number>,<IMEI>,<password>,set,<2021>:1/0*

0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2021:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2021:1*18

14.6.7.1.2 THRESHOLD

This command is used to configure the threshold for accident detection. This configuration is independent of the eco-driving source. The threshold should be configured with the unit rate of change of Force per second.

Example:
$123,860906040263136,aquila123,set,187:3000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,187:1*27

14.6.7.1.3 DEBOUNCE

This command is used to configure Accident Debounce. This can in configured in sub-seconds with one decimal point.

Example:
$123,860906040263136,aquila123,set,181:10*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,181:1*21

14.7 Sleep Mode

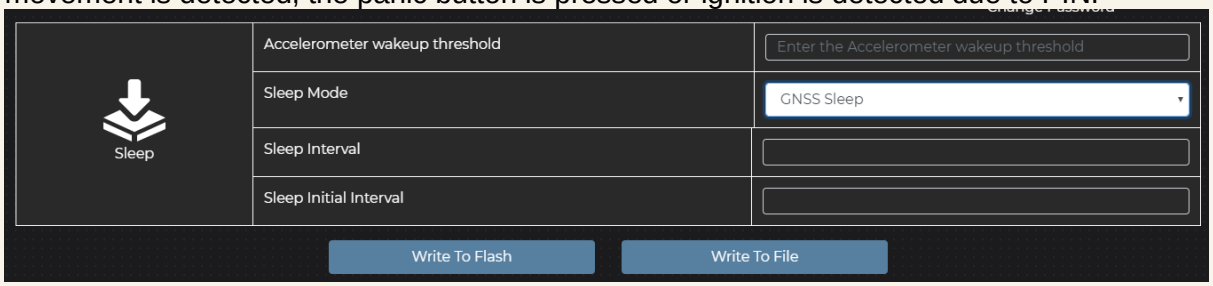
Sleep mode is used to minimize the current consumption of the device during the ignition off state. Below are the different types of sleep modes that can be configured based on the user application. The device will go to sleep mode when the ignition OFF condition. During sleep mode, all the LEDs will be turned off to minimize the current consumption.

- GNSS sleep
- NAP sleep
- Deep sleep

In the case of GNSS sleep, the GNSS section only will put on to the sleep mode. During GNSS sleep the device will wake up in the configured interval. The device will exit the sleep mode in case if the SMS is received, movement is detected, the panic button is pressed or ignition is detected due to PIN.

In the case of NAP sleep, a full processor will be put into sleep mode. During NAP sleep the device will wake up in the configured interval. The device will exit the sleep mode in case if the movement is detected, the panic button is pressed or ignition is detected due to PIN.

In the case of Deep Sleep full processor will be put on to the sleep mode. During NAP sleep the device will not wake up as per interval. The device will exit the sleep mode in case if the movement is detected, the panic button is pressed or ignition is detected due to PIN.



14.7.1 User Commands

14.7.1.1 Sleep Type

The device has an option to configure the different types of sleep mode as per user requirement.

Syntax
$<Random_Number>,<IMEI_Number>,<Device_Password>,set,157:<0/1/3/4>*
0: Disable
1: GNSS Sleep
3: Nap Sleep
4: Deep Sleep

Example:
$123,860906040263136,aquila123,set,157:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,157:0*12

14.7.1.2 Sleep initial Interval

This is the minimum time after which the device will go to the configured sleep mode. This command is used to configure the sleep initial interval time.

Example:
$123,860906040263136,aquila123,set,159:60*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,159:1*2A

14.7.1.3 Sleep interval

This command is used to configure the interval of time-based wake up when the device is in GNSS or NAP Sleep mode.

Example:
$123,860906040263136,aquila123,set,160:300*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,160:1*2E

14.7.1.4 Accelerometer Wake-UP threshold

This command is used to configure the threshold for the accelerometer-based wake up to exit the device from sleep mode.

Example:
$123,860906040263136,aquila123,set,117:100*

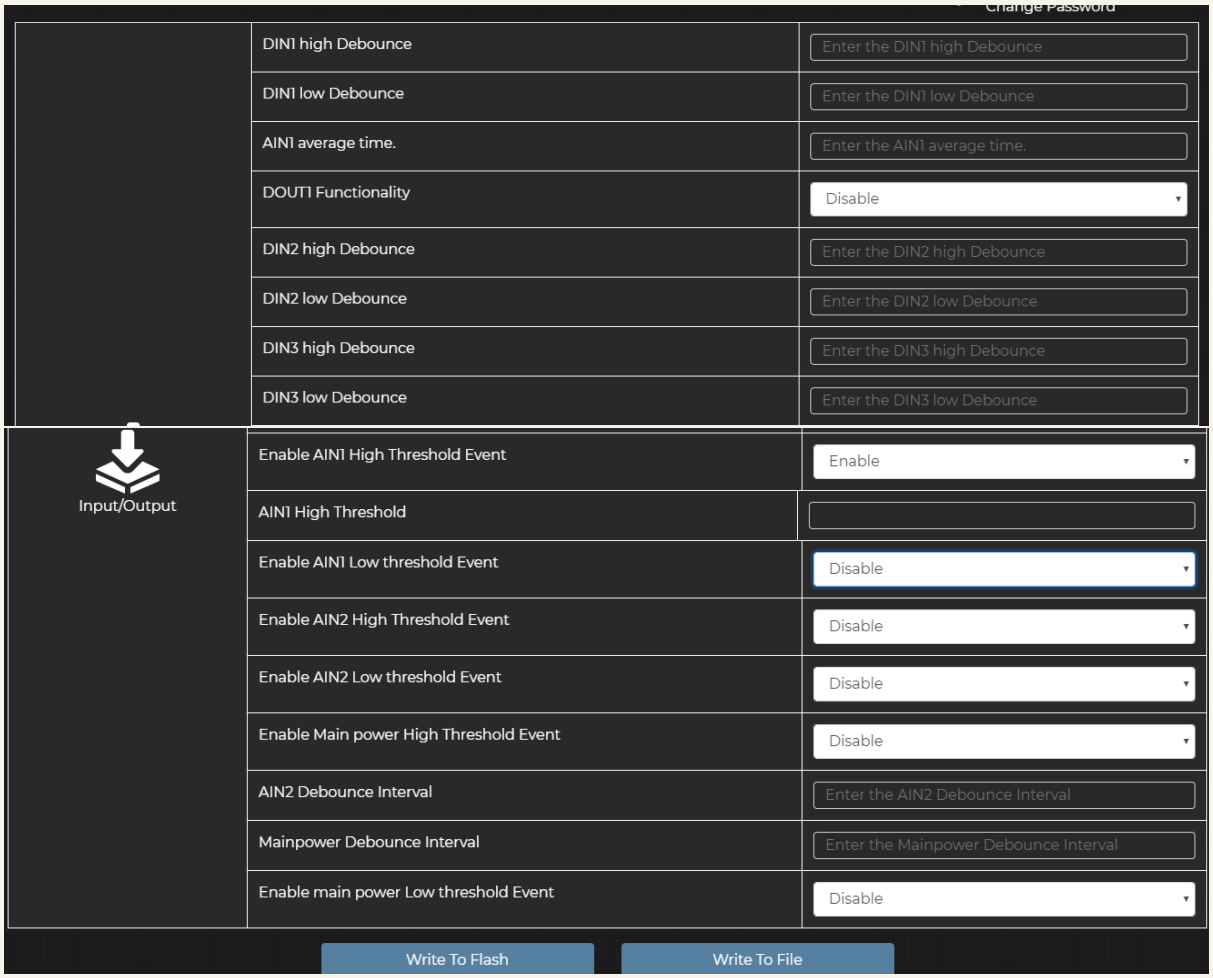
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,117:100*2E

14.8 IO

This section explains the configuration which is related to digital inputs, digital outputs and analogue inputs.

14.8.1 Digital Inputs

The digital inputs can be used to detect the status of the external interfaces. When the status of the interface becomes high, based on the debounce time the device will generate the high event and when the status becomes low, the device will generate the low event.



14.8.1.1 User Commands

14.8.1.1.1 DIN1 HIGH DEBOUNCE

This command is used to configure the debounce time for DIN1 High status. The device will generate a high event when the DIN1 is kept high for debounce time.

Example:
$123,860906040263136,aquila123,set,172:1*
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,172:1*2D

14.8.1.1.2 DIN1 LOW DEBOUNCE

This command is used to configure the debounce time for DIN1 low status. The device will generate a low event when the DIN1 is kept low for debounce time.
Example:
$123,860906040263136,aquila123,set,173:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,173:1*2C

14.8.1.1.3 DIN2 HIGH DEBOUNCE

This command is used to configure the debounce time for DIN2 High status. The device will generate a high event when the DIN2 is kept high for debounce time.

Example:
$123,860906040263136,aquila123,set,197:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,197:1*26

14.8.1.1.4 DIN2 LOW DEBOUNCE

This command is used to configure the debounce time for DIN2 low status. The device will generate a low event when the DIN2 is kept low for debounce time.

Example:
$123,860906040263136,aquila123,set,198:1*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,198:1*29

14.8.1.1.5 DIN3 HIGH DEBOUNCE

This command is used to configure the debounce time for DIN3 High status. The device will generate a high event when the DIN3 is kept high for debounce time.

Example:
$123,860906040263136,aquila123,set,199:1*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,199:1*28

14.8.1.1.6 DIN3 LOW DEBOUNCE

This command is used to configure the debounce time for DIN3 low status. The device will generate a low event when the DIN3 is kept low for debounce time.

Example:
$123,860906040263136,aquila123,set,200:1*

Example:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,200:1*2B

14.8.2 Analog Inputs

The analogue inputs used to read the continuous voltage data from the external interfaces. The device can generate the events based on the voltage thresholds which are configured.

14.8.2.1 User Commands

14.8.2.1.1 AIN1 AVERAGE TIME

This command is used to configure the sampling time of analogue input before it gets updated in the packet. Based on average time, the device starts the accumulation of analogue samples and it will average is based on the number of samples.

Example:
$123,860906040263136,aquila123,set,174:20*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,174:1*2B

14.8.2.1.2 AIN1 HIGH EVENT ENABLE/DISABLE

This command is used to Enable/Disable the AIN1 High Threshold Event.

Syntax:
$<random_number>,<IMEI>,<password>,debug,<2025>:1/0*
0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2025:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2025:1*1C

14.8.2.1.3 AIN1 LOW EVENT ENABLE/DISABLE

This command is used to Enable/Disable AIN1 Low Threshold Event.

Syntax:
$<random_number>,<IMEI>,<password>,debug,<2026>:1/0*
0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2026:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2026:1*1F

14.8.2.1.4 AIN1 HIGH THRESHOLD

This command is used to configure AIN1 High Threshold to generate the event.
Example:
$123,860906040263136,aquila123,set,201:10000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,201:1*2A

14.8.2.1.5 AIN1 LOW THRESHOLD

This command is used to configure AIN1 LowThreshold.
Example:
$123,860906040263136,aquila123,set,202:12000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,202:1*29

14.8.2.1.6 AIN2 AVERAGE TIME

This command is used to configure the sampling time of analogue input before it gets updated in the packet. Based on average time, the device starts the accumulation of analogue samples and it will average is based on the number of samples.

Example:
$123,860906040263136,aquila123,set,269:5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,269:1*24

14.8.2.1.7 AIN2 HIGH EVENT ENABLE/DISABLE

This command is used to Enable/Disable the AIN2 High Threshold Event.

Syntax:
$<random_number>,<IMEI>,<password>,debug,<2027>:1/0*
0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2027:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2027:1*1E

14.8.2.1.8 AIN2 LOW EVENT ENABLE/DISABLE

This command is used to Enable/Disable AIN2 Low Threshold Event.

Syntax:
$<random_number>,<IMEI>,<password>,debug,<2028>:1/0*
0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2028:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2028:1*11

14.8.2.1.9 AIN2 HIGH THRESHOLD

This command is used to configure AIN2 High Threshold.
Example:
$123,860906040263136,aquila123,set,203:11000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,203:1*28

14.8.2.1.10 AIN2 LOW THRESHOLD

This command is used to configure AIN2 Low Threshold.
Example:
$123,860906040263136,aquila123,set,204:13000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,204:1*2F

14.8.2.1.11 MAIN POWER DEBOUNCE

This command is used to configure the debounce interval for the main power voltage for alert generation. For the main power, the single debounce interval will be applicable for both high and low event generations.

Example:
$121,358250330000026,aquila123,set,266:5*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,285:1*2B

14.8.2.1.12 HIGH EVENT ENABLE MAIN POWER

This command is used to enable/disable the main power high event which will be generated when the voltage crosses a high threshold.

Syntax:
$<random_number>,<IMEI>,<password>,set,2047:<1/0>*
0: Disable
1:Enable

Example:
$121,358250330000026,aquila123,set,2047:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2047:1*18

14.8.2.1.13 MAIN POWER HIGH THRESHOLD

The device has the option to configure the threshold for the generation of the main power high event.

Example:
$121,358250330000026,aquila123,set,267:15000*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,267:1*2A

14.8.2.1.14 LOW EVENT ENABLE MAIN POWER

This command is used to enable/disable the main power low event which will be generated when the voltage crosses a low threshold.

Syntax:
$<random_number>,<IMEI>,<password>,set,2048:<1/0>*
0: Disable
1:Enable

Example:
$121,358250330000026,aquila123,set,2048:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,2048:1*17

14.8.2.1.15 MAIN POWER LOW THRESHOLD

The device has the option to configure the threshold for the generation of the main power low event.

Example:
$121,358250330000026,aquila123,set,268:15000*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,268:1*25

14.8.3 Digital Output

There are 2 digital outputs in this device. One digital output is the general-purpose DOUT and another is dedicated to emergency indication. So, the device supports one user configuration digital output. The DOUT 1 can also used as Eco driving indicator based on the configuration.

When the DOUT is being used for immobilizer solution, we recommend using the relay in NC mode to avoid unintentional stoppage of the vehicle during the driving conditions.

14.8.3.1 User Commands

14.8.3.1.1 OPERATE DOUT1

If the DOUT1 functionality is configured as Immobilizer the following section will work as per the commands from the user. Below is the different functionality which it supports.

- Deactivate
- Activate
- Check Speed and Deactivate
- Check Speed and Activate

The 1st two option will be directly visible on the digital output status. The next two option will check of speed (less than 10kmph) to activate or deactivate. If the speed is already less than 10kmph the status will be visible immediately after the command execution.

Syntax:
$<random_number>,<IMEI>,<password>,set,176:<0/1/2/3>*
0: Deactivate
1:Activate
2: Check speed and deactivate
3: Check speed and Activate

Example:
$121,358250330000026,aquila123,set,176:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,176:1*29

14.8.3.1.2 DOUT1 FUNCTIONALITY

DOUT1 can be assigned with different functionality as below.

- Immobilizer
- Eco Driving Indicator

If the DOUT1 is assigned with the functionality of immobilizer, then all the functionality with respect to command key 176 will take effect as per the command.

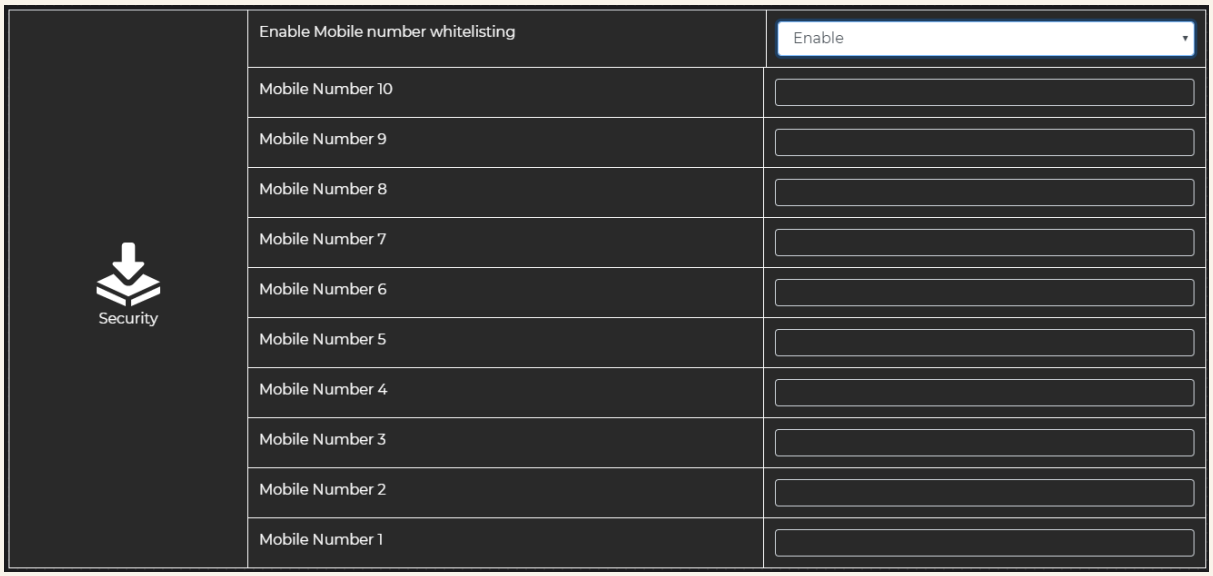
Syntax:
$<random_number>,<IMEI>,<password>,set,175:<0/1>*
0: Immobilizer
1: Eco Driving Indicator

Example:
$121,358250330000026,aquila123,set,175:1*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,175:1*29

14.9 Security

This section explains the security-related functionalities which are supported in the device. The device supports mobile number whitelisting to provide authorized access and unnecessary execution of the commands. Each command that will get executed from the device requires a password for the execution of the command.



14.9.1 Change Password

The device supports the option to change the password of the device. If the user has the current password, then he will be able to change the password of the device. Once the password is changed the changed password should be used for the next set of commands which will be used by the user.

Syntax:
$<random_number>,<IMEI>,<current_password>,set,162:<New_Wassword>*

Example:
$121,358250330000026,aquila123,set,162:aquila321*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,162:1*2C

14.9.2 Mobile Number whitelisting

The device supports the option to whitelist the mobile number for the execution of the commands. If the whitelist is enabled, only the command which is received from a whitelisted mobile will be executed. The device supports up to 10 mobile number those can be whitelisted.

14.9.2.1 User commands

14.9.2.1.1 ENABLE/DISABLE WHITELIST

This command is used to Enable/Disable the Mobile number whitelisting functionality.

Command Syntax:
$<random_number>,<IMEI>,<password>,set,< 2008:1/0*
0:Disable
1:Enable

Example:
$123,860906040263136,aquila123,set,2008:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2008:1*13

14.9.2.1.2 MOBILE NUMBER 1

This command is used to configure the 1st whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,120:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,120:1*2A

14.9.2.1.3 MOBILE NUMBER 2

This command is used to configure the 2nd whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,121:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,121:1*2B

14.9.2.1.4 MOBILE NUMBER 3

This command is used to configure the 3rd whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,122:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,122:1*28

14.9.2.1.5 MOBILE NUMBER 4

This command is used to configure the 4th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,123:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,123:1*29

14.9.2.1.6 MOBILE NUMBER 5

This command is used to configure the 5th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,124:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,124:1*2E

14.9.2.1.7 MOBILE NUMBER 6

This command is used to configure the 6th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,125:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,125:1*2F

14.9.2.1.8 MOBILE NUMBER 7

This command is used to configure the 7th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,126:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,126:1*2C

14.9.2.1.9 MOBILE NUMBER 8

This command is used to configure the 8th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,127:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,127:1*2D

14.9.2.1.10 MOBILE NUMBER 9

This command is used to configure the 9th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,128:+9199000000000*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,128:1*22

14.9.2.1.11 MOBILE NUMBER 10

This command is used to configure the 10th whitelisted mobile number.
Example:
$123,860906040263136,aquila123,set,129:+9199000000000*

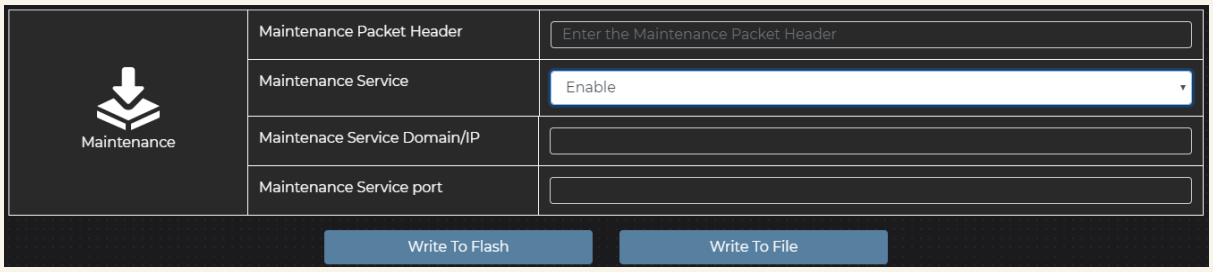
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,129:1*23

14.10 Maintenance

The device supports a separate maintenance service for maintenance of the device along with firmware update over the air. This section explains the maintenance related functionalities that are supported in the device.

14.10.1 Maintenance Service

Maintenance service is the functionality in which the device will communicate to a configured server in a defined interval to check are there any commands or the firmware update pending to be loaded. The communication which is done on maintenance service doesn't contain any location information. The ping contains only the IMEI, timestamp information and current firmware version.



14.10.1.1 User command

14.10.1.1.1 ENABLE/DISABLE MAINTENANCE SERVICE

This command is used to Enable/Disable the Maintenance Service.
Syntax
$<Random_code>,<IMEI_Number>,<password>,set,2010:1/0*
0:Disable
1:Enable

Example:
$123,860906040263136,aquila123,set,2010:0*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2010:1*1A

14.10.1.1.2 MAINTENACE IP/DOMAIN

This command is used to configure the Maintenace server IP/Domain.

Example:
$123,860906040263136,aquila123,set,134:ms1.aquilatrack.com*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,134:1*2F

14.10.1.1.3 MAINTENANCE PORT

This command is used to configure the Maintenace server Port.

Example:
$123,860906040263136,aquila123,set,135:17777*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,135:1*2E

14.10.1.1.4 MAINTENANCE INTERVAL

This command is used to configure the maintenance Server Connect Interval.

Example:
$123,860906040263136,aquila123,set,284:1200*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,284:1*27

14.10.1.1.5 MAINTENANCE PACKET HEADER

This command is used to configure the header of the maintenance Packet which will be sent to the maintenance server.

Example:
$123,860906040263136,aquila123,set,166:CLIENT_1ZF *

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,166:1*28

14.10.1.1.6 TIME OFFSET OF MAINTENANCE SERVICE

This command is used to configure time offset to be used for maintenance ping.

Syntax
$<Random_code>,<IMEI_Number>,<password>,set,2044:0/1*
0: UTC
1: Configured Offset

Example:
$123,860906040263136,aquila123,set,2044:0*

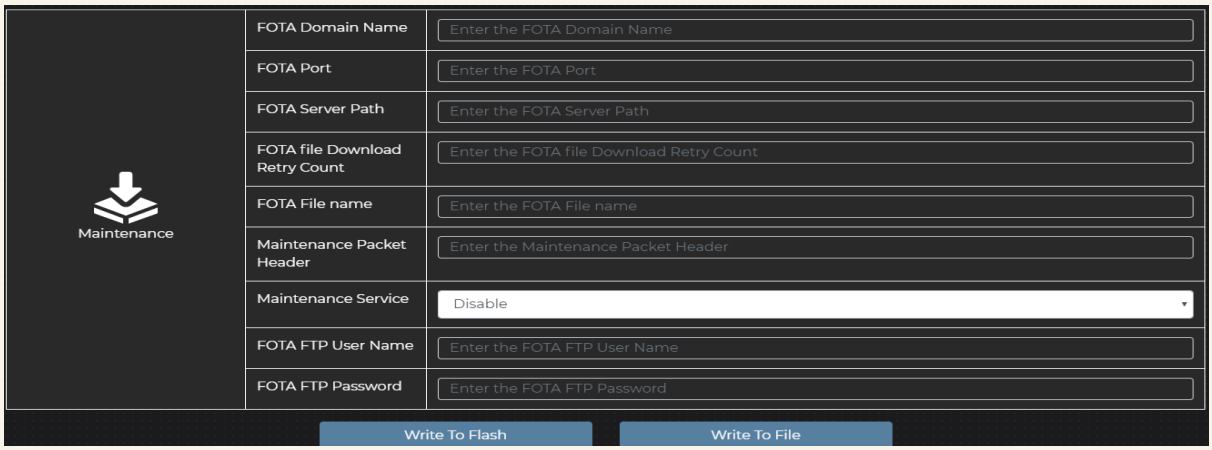
Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2044:1*1B

14.10.2 FOTA

The device supports firmware over the air through which the firmware of the device can be updated remotely. All the sections which have the firmware can be updated over the air. Below are the sections which can be updated in the device. The FOTA follows downloading the firmware by using the FTP service and updating it into the respective section. Whitelisting of the FOTA IP/Domain, Port and FTP service from the mobile network service provider is mandatory to keep the device updated. So don't execute any command after initiation of the FOTA.

- Full Application FOTA
- Controller FOTA

- Once the FOTA download is initiated, all the stored data will be erased from the device memory.
- Any command which is executed after the initiation of the FOTA will not be executed.
- The device will not record any data during the FOTA duration.
- The minimum time which is required to do the Full application FOTA will take 15 minutes.
- The time required for FOTA purely depends on the stable network availability and the retry count which is configured.



14.10.2.1 User Commands

14.10.2.1.1 FOTA IP/DOMAIN NAME

This command is used to configure the FOTA Domain Name.

Example:
$121,358250330000026,aquila123,set,136:ms1.aquilatrack.com*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,136:1*2D

14.10.2.1.2 FOTA PORT

This command is used to configure the FOTA Server Port.

Example:
$121,358250330000026,aquila123,set,137:21*

Response:
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,137:1*2C

14.10.2.1.3 FOTA USER NAME

This command is used to configure the FTP user name which is used to download the file.

Example:
$123,860906040263136,aquila123,set,241:ITBLR*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,241:1*2E

14.10.2.1.4 FOTA PASSWORD

This command is used to configure the FTP password which is used to download the file.

Example:
$123,860906040263136,aquila123,set,242:itblr123*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,242:1*2D

14.10.2.1.5 FOTA FILE PATH

This command is used to configure the FTP path of the binary file.

Example:
$121,358250330000026,aquila123,set,138:/data/Bharat101C/Full/*

Response:_
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,138:1*23

14.10.2.1.6 RETRY COUNT

This command is used to configure the FOTA file Download Retry Count. When the device is downloading the file, if it gets failed due to a data/connectivity related issue, the device will retry to download and update the firmware. This count should be configured sensibly as it will directly affect the no communication time during FOTA

Example:
$121,358250330000026,aquila123,set,139:5*

Response:_
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,139:1*22

14.10.2.1.7 FOTA BINARY FILE

This command is used to configure the FOTA binary file. The device will download the binary file which is configured in the device.

Example:
$121,358250330000026,aquila123,set,140:BHARAT101.bin*

Response:_
$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,140:1*2C

14.10.2.1.8 FOTA INITIATE

This command is used to initiate the FOTA based on the other FOTA configurations.

Syntax
$<Random_code>,<IMEI_Number>,<password>,set,141:0/1/3*
0: No Update
1: Initial Full Application FOTA
3: Initiate Controller FOTA

Example:
$123,860906040263136,aquila123,set,141:0*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,141:1*2D

14.10.2.2 FOTA Procedure

If the FOTA is getting initiated by using SMS or the Bluetooth application it is recommended to initiate the FOTA using set command by multiple key methods. Below is the example command.

Example:

$121,358250330000026,aquila123,set,136:ms1.aquilatrack.com,137:21,241:ITBLR,242:itblr123,138:/data/Bharat101C/Full/,140:BHARAT101.bin,141:1*

Response:

$$CLIENT_1ZF,358250330000026,30,121,0,1,1,set,136:1,137:1,241:1,242:1,138:1,140:1,141:1*12

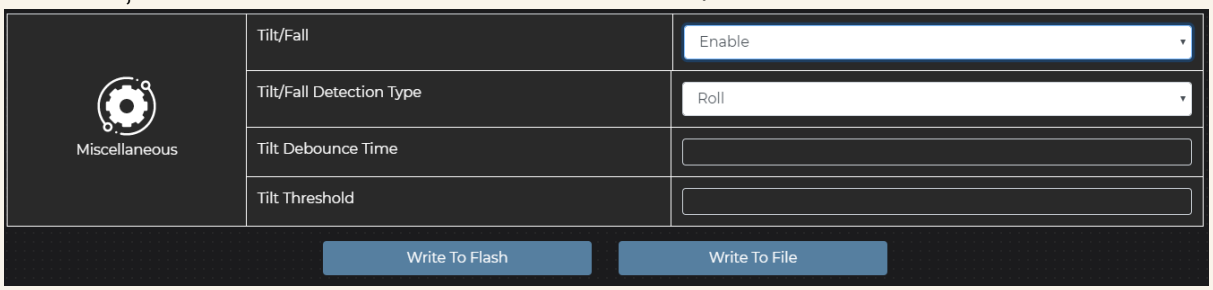
14.11 Miscellaneous

This section explains the miscellaneous functionality which is enabled in the devices.

14.11.1 Tilt/Fall

The device has the option to detect Tilt/fall and generate the event based on the configured thresholds. Auto orientation values should be saved before to detect tilt or fall during the installation. The difference between the current tilt angle and installed angle more than the configured threshold, the device will generate an event.

If the axis is configured as a roll, only the roll axis will be considered for fall and tilt detection. If the axis is configured as pitch only the pitch axis will be used for the detection. If the axis is configured as roll/pitch, the device should be installed horizontally to earth for the proper functioning of the functionality. If the dynamic orientation is selected, the device will monitor all the axis for tilt/fall detection.



14.11.1.1 User command

14.11.1.1.1 ENABLE/DISABLE TILT/FALL

This command is used to Enable/Disable Tilt(Fall)

Syntax:
$<random_number>,<IMEI>,<password>,set,< 2020:1/0*
0: Disable
1: Enable

$123,860906040263136,aquila123,set,2020:1*

Response:

$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2020:1*19

14.11.1.1.2 TILT/FALL DETECTION AXIS

This command is used to configure the tilt/fall detection axis. Based on the detection axis the device will monitor the angular difference and generate the event.
Syntax:
$<random_number>,<IMEI>,<password>,set,< 184:0/1/2*
0: Roll
1: Pitch
2: Dynamic Orientation.

Example:
$123,860906040263136,aquila123,set,184:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,184:1*24

14.11.1.1.3 TILT/FALL THRESHOLD

This command is used to configure the tilt/fall threshold above which the status is considered as tilt/fall.

Example:
$123,860906040263136,aquila123,set,185:40*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,185:1*25

14.11.1.1.4 TILT/FALL DEBOUNCE

This command is used to configure the tilt Debounce time.

Example:
$123,860906040263136,aquila123,set,186:10*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,186:1*26

14.11.1.2 Save Dynamic Orientation Data

This command is used to configure the Save Dynamic Orientation data. After completion of the installation, the vehicle should be parked on the plane and the vehicle should be perpendicular to the earth axis. Then need to send the below command to save the dynamic orientation axis.

Example:
$123,860906040263136,aquila123,debug,111:1*

Response
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,debug,111:1*3B

14.12 UART/RS232

The device has a serial port which can be used as a plug n play port which is can be configurable as per user requirement. The UART can be configured with different modes. Based on the mode the different functionalities are applicable.

- UART
- ECU Connect
- ECU+UART

The subsections explain the different configurations which are supported in the device with respect to UART/RS232 functionality.

14.12.1 User Command

14.12.1.1 UART Mode

The UART port of the device can be enabled/disabled using this command.

Syntax:
$<random_number>,<IMEI>,<password>,set,226:0/1/2/3*
0: Disable
1: UART Mode
2: ECU Mode
3: ECU+UART Mode

$123,860906040263136,aquila123,set,226:1*
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,226:1*2D

14.12.2 UART

The UART is enabled with 2-way communication in which the data can be received from the external accessory and send to the server or the data can be pushed to the device from the server.

The option has been provided to have the communication both in HEX and ASCII mode along with different data acquisition methods. The request/communication to the accessory can be programmed so that any accessory which requires the request to send the information is enabled.

14.12.2.1 User command

14.12.2.1.1 BAUD RATE

This command is used to set the baud rate for UART operation.

Syntax:
$<random_number>,<IMEI>,<password>,set,213:<0/1/2/3/4/5/6/7/8/9/10>*
0:1200
1:2400
2:4800
3:9600

4:19200
5:38400
6:57600
7:115200
8:230400
9:460800
10:921600

Example:
$123,860906040263136,aquila123,set,213:7*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,213:1*2B

14.12.2.1.2 NO OF BITS

This command is used to set the no of bits which is the basic RS232 setting.

Syntax:
$<random_number>,<IMEI>,<password>,set,214:<2/3>*
2:7-bits
3:8-bits

Example:
$123,860906040263136,aquila123,set,214:3*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,214:1*2C

14.12.2.1.3 PARITY BIT

This command is used to set the parity which is basic RS232 settings

Syntax:
$<random_number>,<IMEI>,<password>,set,< 215>:0*
0:None

Example:
$123,860906040263136,aquila123,set,215:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,215:1*2D

14.12.2.1.4 STOP BITS

This command is used to set the stop bit which is basic RS232 settings.

Syntax:
$<random_number>,<IMEI>,<password>,set,216:<0>*
0:1-bit

Example:
$123,860906040263136,aquila123,set,216:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,216:1*2E

14.12.2.1.5 FLOW CONTROL

This command is used to set the Flow control which is basic RS232 settings.

Syntax:
$<random_number>,<IMEI>,<password>,set,217:<0/1>*
0: None
1: Software Flow Control

Example:
$123,860906040263136,aquila123,set,217:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,217:1*2F

14.12.2.1.6 ACCESSORY ID

This command is used to set the accessory ID which will be reflected in each UART packet in accessory ID field.

Syntax:
$<random_number>,<IMEI>,<password>,set,218:<Accessory_ID>*
$123,860906040263136,aquila123,set,218:20*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,218:1*20

14.12.2.1.7 DATA FORMAT

This command is used to set the Data format for UART functionality, this data format can be configurable depends on in which mode data should send from device to server.

When the format is configured as ASCII, the received data will be processed in ASCII and will be sent to the server. When the format is configured as HEX the received data will be processed in hex mode and serial data which is sent to the server will be ASCII converted hex data.

Syntax:
$<random_number>,<IMEI>,<password>,set,219:<0/1>*
0: ASCII
1: HEX

Example:
$123,860906040263136,aquila123,set,219:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,219:1*21

14.12.2.1.8 UART DATA FILTER

This command is used to set the UART data filter to record the received data and send it to the server. The format of the UART data on the server will be based on the data format which is configured.

When the data filter is configured as all data whichever the data received in UART will be pushed directly to the server.

When the data filter is configured as between start and end character, the data which is present between the start and end character will be recorded and sent to the server. If no data available between the character the packet will not be sent to the server.

When the data filter is configured as length, the length of the data which is configured will be sent to the server.

When the data filter is configured as programmable length, the data between the start index and the length after the start index will be recorded and sent to the server.

Syntax:
$<random_number>,<IMEI>,<password>,set,220:<0/1/2/3>*
0: All Data.
1: Between Start and End Character.
2: Length.
3: Programmable Length.

Example:
$123,860906040263136,aquila123,set,220:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,220:1*2B

14.12.2.1.9 START CHARACTER

This command is used to set the start Character, this start character is used when the UART data filter is configured as between start and end character. The value should be the decimal equivalent of the ASCII table.

Syntax:
$<random_number>,<IMEI>,<password>,set,221:<Start_character>*

Example:
$123,860906040263136,aquila123,set,221:65*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,221:1*2A

14.12.2.1.10 END CHARACTER

This command is used to set the End Character, this start character is used when the UART data filter is configured as between start and end character. The value should be the decimal equivalent of the ASCII table.

Syntax:
$<random_number>,<IMEI>,<password>,set,222:<End_character>*

Example:
$123,860906040263136,aquila123,set,222:66*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,222:1*29

14.12.2.1.11 SEND WITH START AND END CHARACTER

This command is used to enable/disable send with start and end character functionality. This configuration will be applicable when the UART data Filter is configured as between start and end character. When this is enabled the start and end character will add with the received UART data in the packet which is sent to the server.

Syntax:
$<random_number>,<IMEI>,<password>,set,2031:<0/1>*
0: Disable
1: Enable

Example:
$123,860906040263136,aquila123,set,2031:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2031:1*1B

14.12.2.1.12 LENGTH OF THE STRING

This command is used to set the length of the string, this is used when the UART data Filter is configured as Length. When the length is configured the data will process up to the configured length in the data string. The maximum length that can be added is 255 bytes/characters based on format configuration.

Syntax:
$<random_number>,<IMEI>,<password>,set,223:<length>*

Example:
$123,860906040263136,aquila123,set,223:66*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,222:1*29

14.12.2.1.13 START INDEX

This command is used to set the start index, this is used when the UART data Filter is configured as programmable length. This is the index of the byte/character in the received string from UART from which the data will be taken.

Syntax:
$<random_number>,<IMEI>,<password>,set,224:<start_index>*

Example:
$123,860906040263136,aquila123,set,224:2*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,224:1*2F

14.12.2.1.14 COUNT AFTER START INDEX

This command is used to set the count after the start index, this is used when the UART data Filter is configured as programmable length.

Syntax:
$<random_number>,<IMEI>,<password>,set,225:<count>*

Example:
$123,860906040263136,aquila123,set,225:254*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,225:1*2E

14.12.2.1.15 PUSH UART

Push UART is a debug command through which any data can be pushed to an external accessory. The value which is added while sending the command will be pushed to the UART port based on the data format which is configured.

Syntax:
$<random_number>,<IMEI>,<password>,debug,128:<value>*

Example:
$123,860906040263136,aquila123,debug,128:abcdfeghijkl*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,debug,128:1*33

14.12.2.1.16 APPEND START STRING

This command is used to Enable/Disable the append start string, if the append start string is enabled, data will receive with the start string in external UART. Otherwise start string that is configured will not be sent to eternal UART.

Syntax:
$<random_number>,<IMEI>,<password>,set,2031:<0/1>*
0: Disable.
1: Enable.

$123,860906040263136,aquila123,set,2031:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2031:1*1B

14.12.2.1.17 START STRING

This command is used to set the start string to get appended in the Push UART command.

Syntax:
$<random_number>,<IMEI>,<password>,set,227:<start string>*

Example:
$123,860906040263136,aquila123,set,227:abcd*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,227:1*1B

14.12.2.1.18 APPEND END STRING

This command is used to Enable/Disable the append end string, if the append end string is enabled, data will receive with the end string in external UART. Otherwise, the end string which is configured will not be sent to eternal UART when the push UART command is received.

Syntax:
$<random_number>,<IMEI>,<password>,set,2032:<0/1>*
0: Disable.
1: Enable.

Example:
$123,860906040263136,aquila123,set,2032:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2032:1*18

14.12.2.1.19 END STRING

This command is used to set the end string. This end string is valid characters/numbers.

Syntax:
$<random_number>,<IMEI>,<password>,set,228:<end_string>*

Example:
$123,860906040263136,aquila123,set,228:efghi*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,227:1*1B

14.12.2.1.20 AUTOMATIC PUSH OVER UART

This command is used to Enable/Disable the automatic push over UART, this functionality is used to send data to external peripherals.

Syntax:
$<random_number>,<IMEI>,<password>,set,<2034>:0/1*
0:Disable.
1:Enable.

Example:
$123,860906040263136,aquila123,set,2034:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2034:1*1E

14.12.2.1.21 ENABLE 1ST PUSH COMMAND

This command is used to Enable/Disable the 1st push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,< 2035>:0/1*
0: Disable.
1: Enable.

Example:
$123,860906040263136,aquila123,set,2035:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2035:1*1F

14.12.2.1.22 1ST PUSH COMMAND

This functionality is used to save the first push command which is used to send for External peripherals.

Syntax:
$<random_number>,<IMEI>,<password>,set,229:<1st push command>*
Example:
$123,860906040263136,aquila123,set,229:PUSH_UART_1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,229:1*22

14.12.2.1.23 DELAY AFTER 1ST PUSH COMMAND

This command is used to configure the delay time to send the command which is saved in the 2nd push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,230:<delay_after_1st_push_Command>*

Example:
$123,860906040263136,aquila123,set,230:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,230:1*2A

14.12.2.1.24 ENABLE 2ND PUSH COMMAND

This command is used to Enable/Disable the 2nd push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,2036:0/1*
0: Disable.
1: Enable.

Exmple:
$123,860906040263136,aquila123,set,2036:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2036:1*1C

14.12.2.1.25 2ND PUSH COMMAND

This functionality is used to save the second push command which is used to send for External peripherals

Syntax:
$<random_number>,<IMEI>,<password>,set,231:<2nd push command>*

Example:
$123,860906040263136,aquila123,set,231:PUSH_UART_2*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,231:1*2B

14.12.2.1.26 DELAY AFTER 2ND PUSH COMMAND

This command is used to configure the delay time to send the command which is saved in the 3rd push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,232:<delay_after_2nd_push_Command>*

Example:
$123,860906040263136,aquila123,set,232:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,232:1*28

14.12.2.1.27 ENABLE 3RD PUSH COMMAND

This command is used to Enable/Disable the 3rd push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,<2037>:0/1*
0: Disable.
1: Enable.

Example:
$123,860906040263136,aquila123,set,2037:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2037:1*1D

14.12.2.1.28 3RD PUSH COMMAND

This functionality is used to save the third push command which is used to send for External peripherals.

Syntax:
$<random_number>,<IMEI>,<password>,set,233:<3rd push command>*

Example:
$123,860906040263136,aquila123,set,233:PUSH_UART_3*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,233:1*29

14.12.2.1.29 DELAY AFTER 3RD PUSH COMMAND

This command is used to configure the delay time to send the command which is saved in the 4th push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,232:<delay_after_3rd_push_Command>*

Example:
$123,860906040263136,aquila123,set,234:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,234:1*2E

14.12.2.1.30 ENABLE 4TH PUSH COMMAND

This command is used to Enable/Disable the 4th push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,2038:0/1*
0: Disable.
1: Enable.

Example:
$123,860906040263136,aquila123,set,2038:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2038:1*12

14.12.2.1.31 4TH PUSH COMMAND

This functionality is used to save the fourth push command which is used to send for External peripherals.

Syntax:
$<random_number>,<IMEI>,<password>,set,235:<4th push command>*

Example:
$123,860906040263136,aquila123,set,235:PUSH_UART_4*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,235:1*2F

14.12.2.1.32 DELAY AFTER 4TH PUSH COMMAND

This command is used to configure the delay time to query data from the next automatic push over the UART command.

Syntax:
$<random_number>,<IMEI>,<password>,set,239:<delay_after_4th_push_Command>*

Example:
$123,860906040263136,aquila123,set,239:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,239:1*23

14.12.2.1.33 ENABLE 5TH PUSH COMMAND

This command is used to Enable/Disable the 5th push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,2039:<0/1>*
0: Disable.
1: Enable.

$123,860906040263136,aquila123,set,2039:1*
Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2039:1*13

14.12.2.1.34 5TH PUSH COMMAND

This functionality is used to save the fifth push command which is used to send for External peripherals.

Syntax:
$<random_number>,<IMEI>,<password>,set,237:<5th push command>*

Example:
$123,860906040263136,aquila123,set,237:PUSH_UART_5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,237:1*2D

14.12.2.1.35 DELAY AFTER 5TH PUSH COMMAND

This command is used to configure the delay time to send the command which is saved in the 1st push command.

Syntax:
$<random_number>,<IMEI>,<password>,set,238:<delay_after_5th_push_Command>*

Example:
$123,860906040263136,aquila123,set,238:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,238:1*22

14.12.2.1.36 RMC OUT

The device has the option to send out the RMC strings which are received from the GNSS section of the device. This command is used to enable/disable the RMC strings on the UART. This will be applicable only if the UART is enabled in the configuration.

Syntax:
$<random_number>,<IMEI>,<password>,set,2061:<0/1>*
0:Disable.
1:Enable.

Example:
$123,860906040263136,aquila123,set,2061:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2061:1*1E

14.12.2.1.37 GGA OUT

The device has the option to send out the GGA strings which are received from the GNSS section of the device. This command is used to enable/disable the GGA strings on the UART. This will be applicable only if the UART is enabled in the configuration.

Syntax:
$<random_number>,<IMEI>,<password>,set,2062:<0/1>*
0: Disable.
1: Enable.

$123,860906040263136,aquila123,set,2062:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2062:1*1D

14.12.2.1.38 REMOVE NON-PRINTABLE BYTES FROM RECEIVED ASCII

The device has the option to remove the non-printable characters from the received UART strings when the ASCII format is enabled. This command is used to enable/disable this functionality.

Syntax:
$<random_number>,<IMEI>,<password>,set,2063:<0/1>*
0:Disable.
1:Enable.

Example:
$123,860906040263136,aquila123,set,2063:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2063:1*1D

14.12.2.1.39 PROTOCOL FORMATTER UART DATA

This functionality is to send the UART packet extended parameters with the serial packet to the primary server based on the configurable value. Each parameter of the 32-bit configuration signified individual fields that should be enabled in the packet protocol. Currently, there are only some parameters that are defined. Below is the table which explains each bit field that is defined for individual parameters. The decimal value of 32-bit parameters should be configured in the command.

e.g.

- If all the defined parameters enabled the decimal equivalent is 4026531840
- If Configurable bit field identifier and Main power voltage is enabled the decimal equivalent is 3221225472
- If Main power voltage, Speed and Ignition status is enabled the decimal equivalent is 1879048192

SL No	Bit Position	Parameter
1	31	Configurable Field Identifier
2	30	Main Power Voltage
3	29	Speed
4	28	Ignition Status
5	27	Reserved
6	26	Reserved
7	25	Reserved
8	24	Reserved
9	23	Reserved

10	22	Reserved
11	21	Reserved
12	20	Reserved
13	19	Reserved
14	18	Reserved
15	17	Reserved
16	16	Reserved
17	15	Reserved
18	14	Reserved
19	13	Reserved
20	12	Reserved
21	11	Reserved
22	10	Reserved
23	9	Reserved
24	8	Reserved
25	7	Reserved
26	6	Reserved
27	5	Reserved
28	4	Reserved
29	3	Reserved
30	2	Reserved
31	1	Reserved
32	0	Reserved

Syntax:
$<random_number>,<IMEI>,<password>,set,154:<decimal_equivalent_bit_field>*

Example:
$$123,358250330003996,aquila123,set,154:4026531840*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,154:1*1D

14.12.3 ECU

The device has the option to connect an Accessory called ECU Connect. This accessory enables the device to read the vehicle data from CAN Bus. The ECU Connect will connect to the UART of the Bharat101 device. As soon as the UART mode is changed to ECU, the device UART baud rate will be changed to either 9600 or 115200 based on the baud rate of the ECU connect is connected. The Bharat101 has the auto capability to detect the baud rate of the ECU connect which is connected.

The ECU connect can read the data in different modes.

- OBD Mode
- 29bit Filtering
- 11bit Filtering

14.12.3.1 User Command

14.12.3.1.1 ECU CONNECT MODE

The device has the option to configure the mode of the ECU connect. Based on the mode, configured, the device will retrieve the data from the accessory and send the data to the server.

Syntax:
$<random_number>,<IMEI>,<password>,set,298:<0/1/2>*
0: OBD.
1: 29 Bit filtering.
2: 11 Bit filtering.

Example:
$123,860906040263136,aquila123,set,298:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,298:1*1D

14.12.3.1.2 CAN BITRATE

The device has the option to configure the bitrate of the CAN but to which the ECU connect is connected.

Syntax:
$<random_number>,<IMEI>,<password>,set,297:<0/1/2/3>*
0: 125kbps.
1: 250kbps.
2: 500kbps.
3:1000kbps.

Example:
$123,860906040263136,aquila123,set,297:2*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,297:1*1D

14.12.3.1.3 ECU CONNECT DATA INTERVAL

The device has the option to configure the data receive interval from the ECU Connect. Based on the configuration the ECU Connect will push the data which is retrieved from the CAN bus to the Bharat101 device. This configuration will be applicable only if the mode is OBD or 29 bit.

Syntax:
$<random_number>,<IMEI>,<password>,set,299:<time_in_seconds>*

Example:
$123,860906040263136,aquila123,set,299:5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,299:1*1D

14.12.3.1.4 CAN PACKET INTERVAL

The device has the option to configure the CAN packet sending interval to the server. Based on this interval the device will record the CAN data and send it to the server.

Syntax:
$<random_number>,<IMEI>,<password>,set,300:<5 seconds >*

Example:
$123,860906040263136,aquila123,set,300:5*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,300:1*1D

14.12.3.1.5 IGNITION DEPENDENCY FOR CAN RECORDING

The device has the option to configure the ignition dependency for CAN Packet recording. If the ignition dependency is enabled, the device will not record the data during ignition off. If the ignition dependency is disabled, the device will record the CAN data irrespective of ignition status.

Syntax:
$<random_number>,<IMEI>,<password>,set,2029:<0/1>*
0: Not dependent on ignition
1: Dependent on ignition

Example:
$123,860906040263136,aquila123,set,2029:0*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,2029:1*1D

14.12.3.1.6 NO OF CAN IDS/PIDS

The device has the option to configure the number of CAN Ids which will be configured in the device. Based on this the CAN Ids should be configured in respective keys.

Syntax:
$<random_number>,<IMEI>,<password>,set,301:16*

Example:

$123,860906040263136,aquila123,set,301:16*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,301:16*1D

14.12.3.2 ODB Mode

In OBD mode the device will query the CAN bus to get the configured PID data. This works based on request-response. When the device is in OBD mode, the device can detect the bit rate automatically.

14.12.3.2.1 USER COMMANDS

14.12.3.2.1.1 OBD PID CONFIGURATION

The device has the option to configure a maximum of 16 PIDs. For each PID separate keys has been defined. Even though all the 16 PIDs are configured, the number of PIDs will be processed based on the “No of CAN IDs/PIDs” configuration. Each key can be configured separately or all 16 PIDs can be configured at once by using multiple key concept. The below table gives the key and respect configuration details.

SL No	Key	Key Specification
1	325	OBD PID 1
2	326	OBD PID 2
3	327	OBD PID 3
4	328	OBD PID 4
5	329	OBD PID 5
6	330	OBD PID 6
7	331	OBD PID 7
8	332	OBD PID 8
9	333	OBD PID 9
10	334	OBD PID 10
11	335	OBD PID 11
12	336	OBD PID 12
13	337	OBD PID 13
14	338	OBD PID 14
15	339	OBD PID 15
16	340	OBD PID 16

Syntax:
$<random_number>,<IMEI>,<password>,set,325:<mode_and_PID>*

Below example is based on configuration of all 16 keys using multiple key configuration concept.

Example:
$108,358250330000083,aquila123,set,325:010C,326:010D,327:0143,328:0105,329:010F,330:0142,331:011F,332:0121,333:0130,334:0131,335:014D,336:014E,337:0111,338:0104,339:015C,340:0161*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,325:1,326:1,327:1,328:1,329:1,330:1,331:1,332:1,333:1,334:1,335:1,336:1,337:1,338:1,339:1,340:1*

14.12.3.3 11bit/29bit Filtering Mode

In 11bit/29bit mode the device will filter the configured CAN Ids from the CAN Bus.

14.12.3.3.1 USER COMMANDS

14.12.3.3.1.1 11BIT/29BIT CAN ID CONFIGURATION

The device has the option to configure a maximum of 16 CAN IDs. For each CAN Ids, separate keys have been defined. Even though all the 16 CAN Ids are configured, the number of CAN Ids will be processed based on the “No of CAN IDs/PIDs” configuration. Each key can be configured separately, or all 16 CAN Ids can be configured at once by using multiple key concept. The CAN ids should be configured in decimal. The below table gives the key and respective configuration details.

SL No	Key	Key Specification
1	305	CAN ID 1
2	306	CAN ID 2
3	307	CAN ID 3
4	308	CAN ID 4
5	309	CAN ID 5
6	310	CAN ID 6
7	311	CAN ID 7
8	312	CAN ID 8
9	313	CAN ID 9
10	314	CAN ID 10
11	315	CAN ID 11
12	316	CAN ID 12
13	317	CAN ID 13
14	318	CAN ID 14
15	319	CAN ID 15
16	320	CAN ID 16

Syntax:
$<random_number>,<IMEI>,<password>,set,305:64*

Below example is based on configuration of all 16 keys using multiple key configuration concept.
Example:
$108,358250330000299,aquila123,set,305:64,306:84,307:66,308:67,309:68,310:69,311:70,312:71,313:72,314:73,315:74,316:75,317:76,318:77,319:78,320:79*

Response:

$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,305:1,306:1,307:1,308:1,309:1,310:1,311:1,312:1,313:1,314:1,315:1,316:1,317:1,318:1,319:1,320:1*1E

14.13 Restricted Vehicle Access

This functionality is used to provide the restricted access of the vehicle to the driver. There are 2 types of access control functionalities that can be configured in the device.

- Identification
- Authorisation.

14.13.1 Identification

In case of identification of the driver who is driving the vehicle, to force use RFID cards to swipe before starting the vehicle, this functionality has been implemented. As soon as the device receives packet it will mobilize the vehicle so that the vehicle will be able to start.

14.13.2 Authorisation

In case of authorisation, only the valid RFID cards from the list of authorised cards will be used for mobilizing the vehicle. As soon as the RFID cards has been received, it will check whether the list which is provided matches with the received driver id.

14.13.3 User Commands

14.13.3.1 Configure Authorisation Functionality.

This command is used to configure the restricted access type.

Syntax:
$<random_number>,<IMEI>,<password>,set,11100:<0/1/2>*
0: Disable
1: Identification
2: Authorisation

Example:
$123,<IMEI>,<device_password>,set,111000:2*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,set,11100:1*12

14.13.3.2 Authorisation ID Configuration

This command is used to configure the authorisation Ids in the device. There are 50 IDs those can be configured. (Key 20087 to 20136)

Syntax:
$<random_number>,<IMEI>,<password>,set,20087:<Authorised ID 1>*

- Authorised ID 1 is RFID data
  Ex:-45F253D6

Example:

$123,<IMEI>,<password>,set,20087:45F253D6*

Response:
$$CLIENT_1ZF,XXXXXXXXXXXXXXXX,30,123,0,1,1,set,20087:1*32

15 Debug Commands

The debug commands are used to debug the device remotely. The status will not be stored in the memory after the restart. The debug commands can be executed from all the channels unless it is specified. This command also explains the command which depends on the registration of the vehicles at common layers.

15.1 Switch Network

It is used to switch from the primary network to a secondary network or vice-versa. If the current profile is 1st profile, then the device will switch it to the second profile and if it is in secondary then will switch it to the primary profile.

15.1.1 User command

This command is used to switch the network.

Syntax:
$<random_number>,<IMEI>,<password>,debug,115:1*

Example:
$123,860906040263136,aquila123,debug,115:1*

Response:
$$CLIENT_1ZF,860906040263136,30,123,0,1,1,debug,115:1*19

15.2 Forced Restart

The device can be forcefully restarted by using commands. The different section of the devices can be forcefully restarted by using the forced restart command.

- Forced Soft Restart
- Forced Hard Restart
- Force Maintenance Restart
- Forced Controller Restart
- Forced GNSS module restart

15.2.1 User Command

This debug command is used to restart the different section of the device along with full device restarts.

Syntax:
$<random_number>,<IMEI>,<password>,debug,100:<0/1/2/3/4>*
0: Forced Soft Restart

1: Forced Hard Restart
2: Forced Maintenance Restart
3: Forced Controller Restart
4: Forced GNSS restart

Example:
$123,358250330000513,aquila123,debug,100:0*

Response:
$$CLIENT_1ZF,358250330000513,30,123,0,1,1,debug,100:1*36

15.3 Activation Command

This command is used to get the activation details of the device by sending through SMS. This command will work only on the SMS channel.

Syntax:
ACTV,Random_Code,Reply SMS Gateway number

Example:
ACTV,123456,+919538352976

REPLY FORMAT:
<ACTVR>,<Randomcode>,<VendorID>,<Firmwareversion>,<IMEI>,<AlertID>,<Latitude>,<direction>,<Longitude>,<Direction>,<GPSfix>,<DateandTime>,<Heading>,<Speed>,<GSMStrength>,<CountryCode(MCC)>,<NetworkCode(MNC)>,<LAC>,<MainPower>,<IGNStatus>,<Battery Voltage>,<FrameNumber>,<Ignitionon/offID>

ACTVR,123456,iTri,1_36T0,868728032125970,01,000.00000000,N,000.00000000,E,0,060420 20 165353,000.00,00.0,10,404,0071,32c8,0,1,00.0,000001,IN

15.4 HCHK Command

This command is used to get the Health details of the device by sending through SMS. This command will work only on the SMS channel.
Syntax:
HCHK,Random_Code,Reply SMS Gateway number

Example:
HCHK,123456,+919538352976

Reply Format
HCHKR,<Randomcode>,<VendorID>,<Firmwareversion>,<IMEI>,<AlertID>,<Latitude>,<direction>,<Longitude>,<Direction>,<GPSfix>,<DateandTime>,<Heading>,<Speed>,<GSMStrength>,<CountryCode(MCC)>,<NetworkCode(MNC)>,<LAC>,<MainPower>,<IGNStatus>,<BatteryVoltage>,<FrameNumber>,<Ignitionon/offID>

HCHKR,123456,iTri,1_36T0,868728032125970,01,000.00000000,N,000.00000000,E,0,060420 20 170118,000.00,00.0,11,404,0071,32c8,0,1,00.0,000002,IN

15.5 Get All

The device supports an option to get all the configurations that are supported in the device along with the status of the configuration values. This command is supported only through serial mode.

Example:
$123,860906040263136,aquila123,debug,119:1*

Response:
101:10.000000
102:2
103:taisysnet
104:0
105:0
106:0
107:3600
108:ms1.aquilatrack.com
109:pool.ntp.org
110:0
111:103.19.89.88
112:17777
114:device.vltsecurity.com
115:8082
116:60
117:100
118:0
119:0
120:+910000000000
121:+910000000000
122:+910000000000
123:+910000000000
124:+910000000000
125:+910000000000
126:+910000000000
127:+910000000000
128:+910000000000
129:+910000000000
130:10
131:1000
132:100
133:300
134:ms1.aquilatrack.com
135:17777
136:ms1.aquilatrack.com
137:21
138:/data/Bharat101C/
139:2

140:image.bin
141:0
142:1
143:20
144:1
145:5
147:3
148:10
149:10
150:80
151:5
152:0
153:0
155:0
156:0
157:0
158:2
159:120
160:600
161:12000
163:Header
164:Header
165:CLIENT_1ZF
166:CLIENT_1ZF
167:0
168:12000
169:1
170:5
171:10
172:1
173:1
174:1
175:0
176:0
177:5
178:20
179:12000
180:12700
181:10
182:5
183:200
184:2
185:40
186:10
187:3000
189:1.000000
190:10
191:Airtel
192:taisysnet
193:BSNL
194:bsnlnet
195:device.vltsecurity.com

196:1090
197:1
198:1
199:1
200:1
201:8000
202:3000
203:8000
204:3000
205:5
206:60
207:5
208:7200
209:1200
210:+910000000000
211:KA1234
212:iTriangle1
213:3
214:3
215:0
216:1
217:0
218:1
219:1
220:1
221:2
222:3
236:+910000000000
241:ITBLR
242:itblr123
243:Header
244:600
245:60.243.245.181
246:17018
251:1
252:1
253:BH101C_
260:0.100000
261:0.100000
262:0.100000
263:1
264:1
265:1
266:5
267:8000
268:3000
269:5
270:40
271:10
284:86400
285:1
290:0

2000:0
2001:0
2002:0
2003:0
2004:0
2005:0
2006:1
2007:0
2008:0
2009:0
2010:1
2011:0
2012:0
2013:0
2014:0
2015:0
2016:0
2017:0
2018:1
2019:0
2020:0
2021:0
2022:1
2023:0
2024:0
2025:0
2026:0
2027:0
2028:0
2029:0
2030:0
2031:0
2032:0
2033:0
2034:0
2035:0
2036:1
2037:0
2038:0
2039:0
2040:1
2041:0
2042:0
2043:0
2044:0
2045:0
2046:0
2047:0
2048:0
2049:0
2050:0
2051:0

2052:0
2053:0
2054:0
2055:0
2056:0
2057:0
2058:0
2059:0
2060:0
2061:0
2062:0
2063:0
2064:0
2065:0
2066:0
2067:0
2068:0
2069:0
2070:0
2071:0
2072:0
2073:0
2074:0
2075:0
2076:0
2077:0
2078:0
2079:0
2080:0
2081:0
2082:0
2083:0
2084:0
2085:0
2086:0
2087:0
2088:0
2089:0
2090:0
2091:0
2092:0
2093:0
2094:0
2095:0
2096:0

16 Diagnostic commands:

A diagnostic command is used to get the general status information of the device over SMS.

16.1 General Device Information

The device supports diagnostic command which can be used to get the general device information like configuration and the status of the communication.

Syntax:
$<random_number>,<IMEI>,<password>,diag,100*

Example:
$123,860906040263136,aquila123,diag,100*

Format of the return message from the device:
$diag|<Serial_Number/IMEI>|<Client_ID>|<Firmware_Version>|<Access_Point_Name>|<1IP/Domain>|<Destination_port>|<2IP/Domain>|<Destination_port>|<3IP/Domain>|<Destination_port>|<Time_Offset>|<GSM_Status>|<current_odometer/delta>|<GNSS_Fix_Status>|<Admin_Number1>|<IGN_On_Interval_SRV1>|<Ignition_off_Interval_SRV1>|<IGN_On_Interval_SRV2>|<Ignition_off_Interval_SRV2>|<Server1_Socket_Status>|<Server2_Socket_Status>|<Server3_Socket_Status>*<Checksum>

Example:
$diag|860906040263136|100|iTriangle1|BHARAT101C_0211|bsnlnet|106.51.85.249|17001|106.51.85.249|17002|106.51.85.249|17003|0|0|379.114136|1|+910000000000|10|300|10|300|300|300|300|*11

<GSM_STATUS>
1: GPRS_Registation Failed
2: GPRS Registration is Success
3: PDP Activation Failed
4: PDP Activation is Successful.

<GNSS_Fix_Status >
0: No Fix
1: Valid GNSS Fix
2: No String from GNSS
<Socket_Status >
1: Domain Name 1 Wrong
2: Domain Name 1 Right
3: SOC Create Fail
4: SOC Create Pass
5: SOC Connect Fail
6: SOC Packet send Success

16.2 Get Location link

The device supports the option to get the location link from the device over SMS. Once the link is received, just by clicking on the link the vehicle location can be viewed on the google map.

Syntax:
$<random_number>,<IMEI>,<password>,diag,101*

Example:
$123,860906040263136,aquila123,diag,101*

Response:
http://maps.google.com/maps?q=12.981880,77.548317

16.3 IO Status

The device supports getting IO information over SMS. This command shows all the interface status over the SMS to the user.

Syntax:
$<random_number>,<IMEI>,<password>,diag,102*

Example:
$123,860906040263136,aquila123,diag,102*

Response:
$diag|<IMEI_NO>|<key>|<0=IGN_Off/1=IGN_On>|<0=DIN1_Low/1=DIN1_High>|<0=DIN2_Low/1=DIN2_High>|<0=DIN3_Low/1=DIN3_High>|<0=DOUT1_Inactive/1=DOUT1_Active>|<0=DOUT2_Inactive/1=DOUT2_Active>|<0=Emergency_Off/1=Emergency_On>|<0=Main_power_Disconnected/1=Main_Power_Connected>|<Main_Power_Voltage_in_mV>|<Internal_Battery_Voltage_in_mV>|<Analog_Input1_Voltage_in_mV>|<Analog_Input2_Voltage_in_mV>|<1=Case_Open/0=Case_Closed>|<Last_Received_RS232_Data_Timestamp>*<Checksum>

Example:
$diag|860906040263136|102|1|0|0|0|0|1|0|1|12|3926|21|180|0*7A

16.4 Get all firmware versions

The device supports the option to get all the firmware versions of individual sections of the device. Below is the command details.

Syntax:
$<random_number>,<IMEI>,<password>,diag,103*

Example:
$123,860906040263136,aquila123,diag,103*

Response:
$diag|<IMEI_NO>|<key>|<hardware_Version>|<Core_Version>|<Fw_Version>|<Controller_Fw_Version>|<GNSS_Fw_Version>*<Checksum>

Example:
$diag|358250330000299|103|VER-01_REV02|CORE|BHARAT101C_0222|CONTROLLER_VER|GNSS_FWVER*3C

16.5 Get GNSS Information

The device supports getting the current GNSS related information directly from the device. These commands work only on the SMS channel.

Syntax:
$<random_number>,<IMEI>,<password>,diag,104*

Example:
$123,860906040263136,aquila123,diag,104*

Response:
$diag|<IMEI_NO>|key|<1=GNSS_Module_On/0=GNSS_Module_Off>|<1=GNSS_Fixed/0=GNSS_not_Fixed>|<TTFF_in_sec>|<Latitude>|<longitude>|<Speed>|<Altitude>|<Heading>|<HDOP>|<PDOP>|<GPS_Satellites_Count>|<IRNSS_Satellite_Count>|<Total_Visible_Satellite>|<Total_Used_Satellite>*<Checksum>

Example:
$diag|860906040263136|104|1|1|659|12.981510|77.548561|0|1050|192|4.40|7.10|18|7|25|7*361

16.6 Get GSM Information

The device supports getting the current GSM and connectivity related information directly from the device. These commands work only on the SMS channel.

Syntax:
$<random_number>,<IMEI>,<password>,diag,<command Key>*

Example:
$123,860906040263136,aquila123,diag,105*
Response:
$diag|<IMEI_NO>|<key>|<1=Radio_On/0=Radio_Off>|<SIM_Status>|<SIM_Profile_Name>|<CCID>|<IMSI>|<CREG_Status>>|<CGREG_Status>|<Signal_Strength>|<Operator_Code>|<1=Bearer_Activated/0=Bearer_Deactivated>|<0=Sokect1_Disconnected/1=Socket1_Connected>|<0=Sokect2_Disconnected/1=Socket2_Connected>|<0=Sokect3_Disconnected/1=Socket3_Connected>|<Last_Record_Socket1_Sent_Timestamp>|<Last_Record_Socket2_Sent_Timestamp>|<Last_Record_Socket3_Sent_Timestamp>*Checksum>

Example:
$diag|860906040263136|105|0|1|0|0|0|1|1|29|0|0|1|1|0|20012021123137|20012021123136|01012070000000*0

16.7 deviceinfo

The above command is used to get the device info like IMEI, Serial Number and Firmware version from the device.

Syntax: deviceinfo
Example:
deviceinfo
Response:
$INFO|<IMEI_NO>|<Event_Code>|<Serial_No>|<HW_Model>|<fw_version>*<Checksum>

Example: $INFO|358250330000026|30|31071993|BH101C|BHARAT101C_0207*19

17 Bluetooth Application

The device supports the configuration and debugging of the devices using the Bluetooth channel. The device supports Bluetooth App which is supplied by the iTriangle. Below are the steps which will be followed during the configuration.

- Launch the App.
- Login using user credentials.
- Scan the device and connect.
- Enter the device password.
- Do the required Operation.
- Disconnect the Device/Tear down the device.

17.1 Mobile Requirements

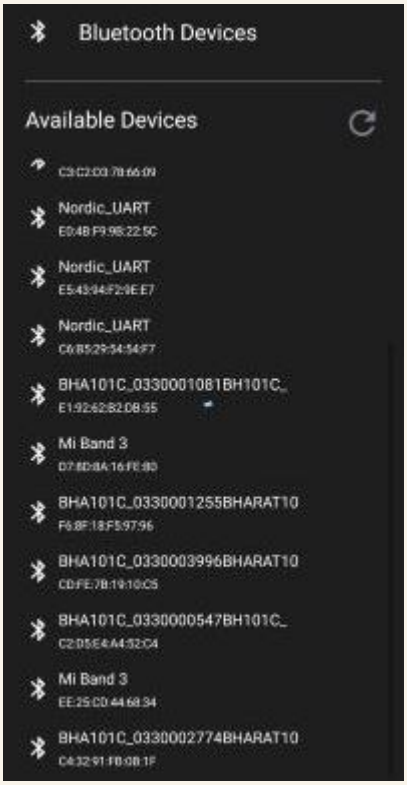
- Minimum Android Version: 6.0 (Marshmallow)
- The mobile should have internet connectivity to log in to the Mobile App.

17.2 Log in to the Mobile App

- User login page takes userID and password from user to login.

- After entering two inputs click on the login button
- After a successful login, the application moves to Device Scan Page

17.3 Scan And Connect



- This Page requests permission for BLE scan (Only 1st Time User)
- Allow permission to use this mobile application
- Click on enable Bluetooth
- After enabling, Bluetooth scan starts and the user can see a list of devices
- Click on the respective device to connect. (BHARAT101C_<Last 10 Digits of IMIE>)
- Top right refresh icon to rescan devices.
- If the connection is successful app opens the device password verification page

17.4 Password Verification

- Enter the device password and click on the verify button.
- Wait until the device responds.
- If the password is wrong, the app shows an Invalid device password message.
- If the password is correct, the app opens the home screen as shown below.

17.5 Application Features

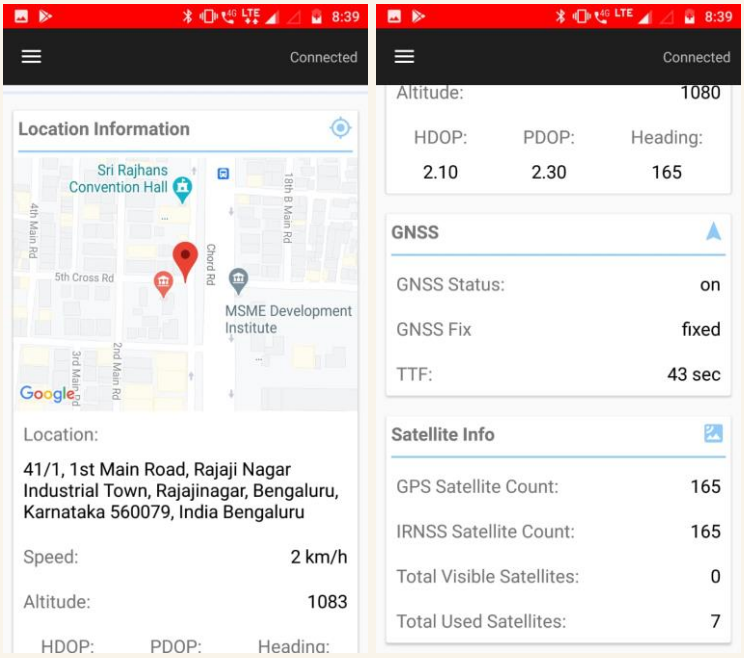
- The application supports reading of the device information, configuration of the device and debugging of the device.

17.6 Device Info

Below is the information which will be displayed on this page.

- Model of the Device
- IMEI number of the Device
- The firmware version of the Device.

17.7 GNSS Info

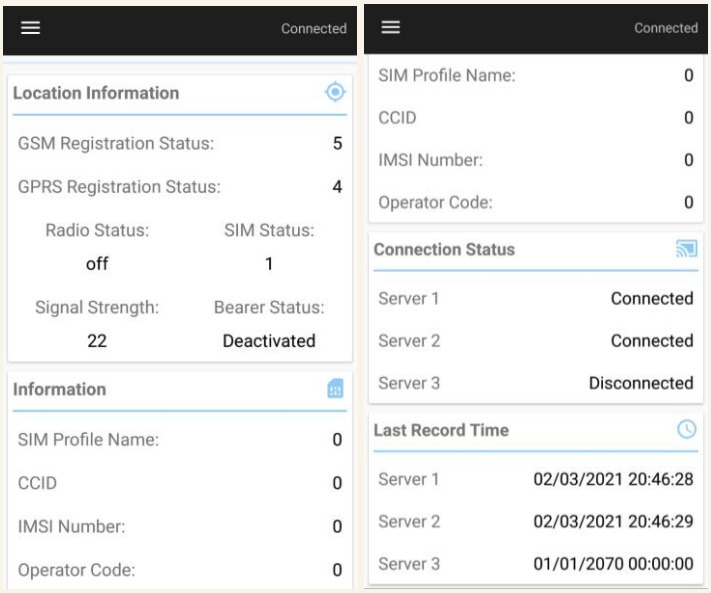


The GNSS info page will show the GNSS related information. This page includes below individual data.

- GNSS Module Status
- GNSS Fix Status
- TTFF
- Location
- Speed
- Altitude
- Heading
- HDOP
- PDOP

- GPS Visible Satellite Count
- IRNSS Visible Satellite Count
- Total Visible Satellites
- Total Used Satellites

17.8 GSM Info



The GSM info page will show the GSM related information. This page includes below individual data.

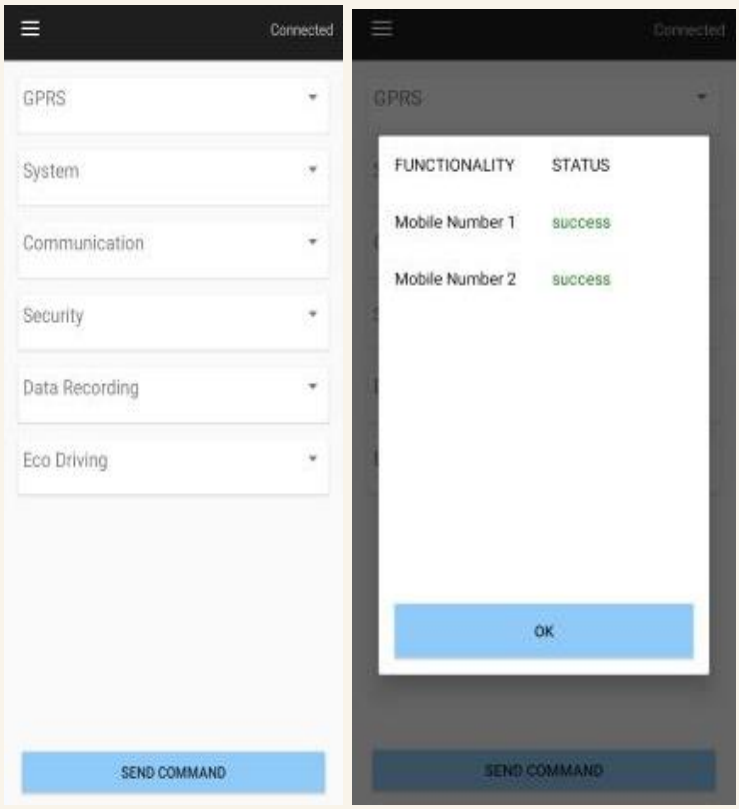
- Radio Status
- SIM Status
- SIM Profile Name
- CCID
- IMSI Number
- GSM Registration Status
- GPRS Registration Status
- Signal Strength
- Operator Code
- Bearer Status
- Server1 Connection Status
- Server2 Connection Status
- Server3 Connection Status
- Last Record Sent Time to server 1
- Last Record Sent Time to Server 2
- Last Record Sent Time to Server 3

17.9 Input/Output

The IO info page will show the IO related information. This page includes below individual data.

- Ignition Status
- DIN1 Status
- DIN2 Status
- DIN3 Status
- DOUT1 Status
- DOUT2 Status
- Emergency Status
- Main Power Status
- Main Power Voltage
- Internal Battery Voltage
- Analog Input 1 Voltage
- Analog Input 2 Voltage
- Case Open Status
- Last Received RS232 Data

17.10 Configuration



This page is used to configure the basic minimum configuration of the device. Below is the process involved in the configuration of the device.

- Select the Command.
- Load the Values
- Click on SEND COMMAND.
- Command execution status will be shown in the popup.

17.11 Debug Mode

The debug page has 3 modes.

- Quick Debug
- Manual
- Logging

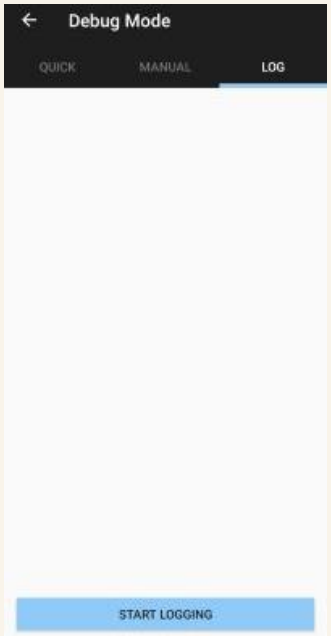
17.12 Quick Mode

This page will give the option to pre-configure buttons with the required command. On clicking on the button the pre-configured text will be sent to the device over the Bluetooth channel.

17.13 Manual Mode

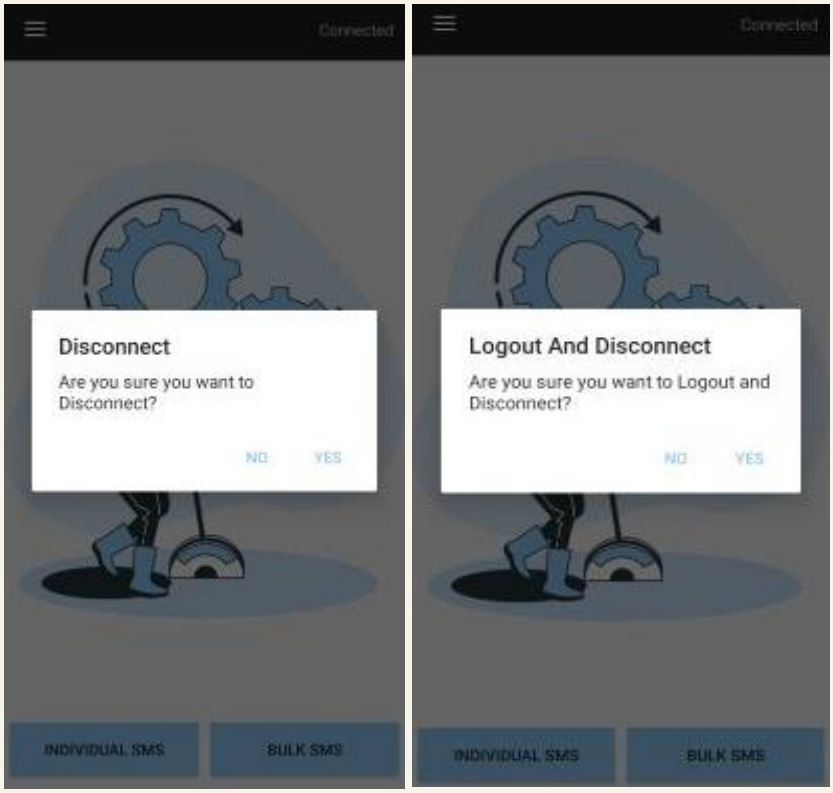
In manual mode, the user can manually send a command to the device and the response will be shown on the screen.

17.14 Logging Mode



When the logging mode tab is opened, and click on the start logging button to start the debug logs on the screen.

17.14.1 Logout/Disconnect



- The navigation menu has a disconnect and a logout option.
- On click of disconnect user will disconnects from the device (not logged out).
- With on click of logout, the user will disconnect from the device and logged out from the app.

18 Device installation

The devices can be installed in any type of vehicle which complies with the power supply range of the device. The device should be installed by a technician who has trained on the installation of the Bharat101 device.

It is highly recommended that the installation of the BHARAT 101 device and its accessories must be done by a trained technician only.

Handle the device with care and do not place any heavyweights on the device. There is a Li-ion battery inside.

It is recommended that the BHARAT 101 device not be installed in places prone to water and ain. Remove BHARAT 101 device during servicing to avoid accidental water splash and high voltages.

18.1 Preparation for the Installation

This section explains the steps which should be followed before installing the device.

18.1.1 Pre Installation notes

- The latest vehicle has closed-loop systems which are controlled by different ECUs. Due to this, there are might be the restriction to get the devices installed in the vehicles. So consult the manufacturer or the local dealers to get specific guidelines for the installation. The guidelines might include where to install the device, specific connector, connection to use for Telematics devices. Not following the instructions from the manufacturer/ dealer might lead to malfunction of the vehicle's electronic/electrical systems even void the warranty of the vehicle.
- Check for the telematics gateway connector option in the latest vehicles from the dealer/manufacturer for the safe connectivity of the device to the vehicle.
- Make sure that you will have the latest and updated user manual for the installation as the documentation might change frequency which might impact the installation method.

18.1.2 Installation Safety

This section explains the installation safety for both vehicle and the technician.

- Disconnect the vehicle battery terminals from the vehicle during the installation. Working with live wires will be dangerous as there are chances of wires getting short during the installation process.

- Installation in the vehicles which has the closed-loop system might lead to unexpected behaviour of the system. Please consult the dealer or the manufacturer and the information before in hand.
- Make sure that ignition status is in off condition before start the installation of the device in the vehicle.

18.1.3 Tools Required

Below is the list of tools that are required for the proper installation of the devices in the vehicle.

- Multimeter
- Crimping Tools for snap-fits and wiring lugs.
- Pliers
- Wire Strippers
- Wire Cutters
- Insulation tapes.
- Knife
- Cable Ties
- Conduit pipes (Based on the routing of wires)

18.2 Installation Instructions

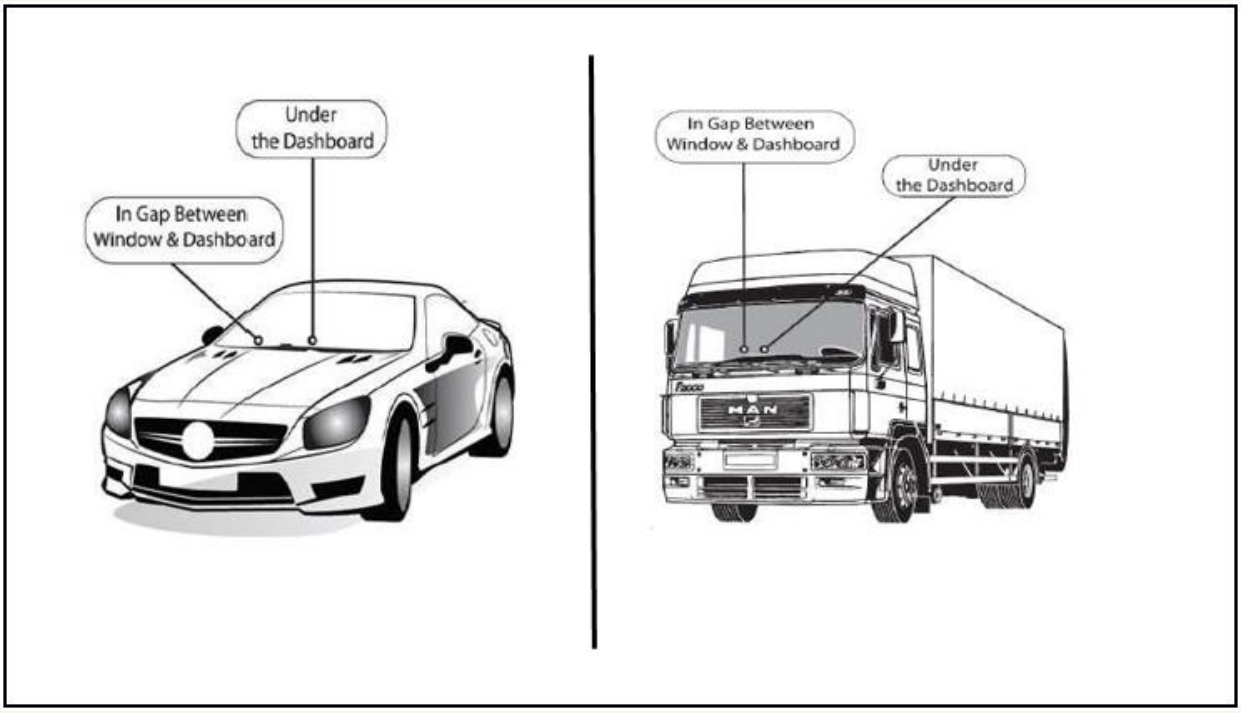
18.2.1 Identification of Installation location.

Below are points that should be taken into considerations for identifying the location of the installation in the vehicle.

- The device contains the radio module which will transmit and receive the GSM and GNSS signals.
- The device will not operate in extreme heat conditions. Kindly check the specifications for understanding the operating condition of the device.
- The device requires covert installation where the devices should not be made easily accessible so that the thieves can find the devices and remove them.
- The metal over and around the device will affect the GNSS signals which will be received from the device.
- The device should have maximum visibility to the sky avoiding metal parts.
- The device should not be installed near the moving parts as it may affect the device performance and might damage the device.
- There should not be any ECUs 20 cms around the location chosen.
- The windshield heaters affect the reception of the GNSS signals.

18.2.1.1 Recommended Install Locations

- Inside/Decide/Above the glove box.
- Behind the dashboard.
- Top of the dashboard, if the dashboard is made up of metal.



18.2.1.2 Location to Avoid

- Close to Airbags
- Under the Roof of the Vehicle
- Inside AC Duct
- Engine Compartment.
- Boot
- Near the Fuel Tank
- Near the Speakers
- Outside the Passenger Compartment of the Vehicle
- Near Antenna or the Radio Transmitters.

18.2.2 The orientation of the Device

The device has an internal GNSS antenna and the accuracy of the location which is received is purely dependant on the quality of the signal received. So the orientation of the device has a greater impact on the GNSS signal reception. It is mandatory to keep the LED side of the device facing the sky, unobstructed by any metal. In any case, there should be no metal on top of the device for up to 30-40 cm around the device. It is not recommended to tilt more than 50 degrees to the horizontal axis to avoid the poor reception of the GNSS signals.

The orientation of the device with respect to the driving axis plays a major role in the case if the eco-driving source is selected as Accelerometer. Even though there is an option to configure the eco-driving axis, the device should be mounted parallel to the vehicle's horizontal axis.

18.2.3 Identify the Wires

Identify the individual wires from the vehicle circuitry as per dealer or the manufacturer guidelines and connect the respective wires of the device to the vehicle based on user requirement.

18.3 Best Installation Practices

The below section explains the best installation practices for a stable and risk-free installation.

- Make sure that all the accessories are in working condition before you start the installation of the devices in the vehicle. If any malfunction is observed immediately inform the authorised persons from the customer/ vehicle owner side.
- For stripping of the wires in the vehicle always use the wire stripper instead of a cutter. The cutters might damage the conductors which are covered by insulation.
- Always isolate the open wires by using insulation tape.
- Don't strip all the required wires at once and start connecting them to the device. Strip the second wire only after complete insulation of the connection which is getting handled currently.
- Use only existing wire ducts and opening to route the wires.
- Make sure that wires are protected by grommets and plastic conduit pipes to avoid damage to the wires.
- Use only a multimeter to test the voltage on the wires in the vehicle.
- While checking the voltage t the wires, don't expose or pinch the wires. Check the voltage at the end of the connector to avoid unnecessary damage to the wires.
- Do not insert the multimeter probe into the female connector as it will widen it and might create loose contact between the male terminals.
- To connect the -ve wire, connect a lug properly crimped to the negative wire of the device and should be screwed on to the chassis using an existing screw and fasten it so that there will be stable connectivity.
- All wires and cables should be hidden for a neat and clean installation.
- To attach the device to its location, use a thick, two-sided, adhesive tape, between the device and the fixed support and then use two plastic straps (cable ties) to secure the device to its location. Make sure the device is well positioned and will not become loose. If there is an option to use the screw/nut and bolt kindly use it for firm installation of the device.
- After completion of the installation, make sure that all the accessories are working as per the same condition which is noted before starting the installation.
- When the installation is finished clean the vehicle and put back all the items you removed into their original positions, using the original screws and connectors.
- Test the connection with all the required servers to ensure proper operation of the device.
- When the DOUT is being used for immobilizer solution, kindly make sure that relay is being used in NC mode to avoid unintentional stoppage of the vehicle during the driving conditions.